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ORIGINAL ARTICLE

Enhancing water quality index prediction accuracy in Mranti lake and rivers in Malaysia using regression forest model

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Abstract

The economic and environmental commodity of water quality has a substantial impact on the welfare of the public and the viability of ecosystem of a country. This research aims to improve the accuracy of water quality index (WQI) prediction by using machine learning techniques based on Malaysian standard. The study is carried out using a historical dataset that includes 11,065 samples from different places by the Malaysian Department of Environment (DOE). The data contains several important water quality parameters, including pH, dissolved oxygen, total suspended solids, biological oxygen demand, chemical oxygen demand, and ammoniacal nitrogen concentration. We develop a novel model, time series regression forest (TSRF) that is based on regression and random forest theory to enhance the WQI prediction. TSRF is compared with other machine learning models: decision trees (DT), ridge regression (RR), artificial neural networks (ANN), extra tree regressor (ETR), random forest (RF), autoregressive integrated moving average (ARIMA), DT-ARIMA and RF-ARIMA models. TSRF achieves the lowest mean square error (MSE) of 1.178 and the highest R^2 of 0.968, surpassing DT–ARIMA (MSE = 2.526, R^2 = 0.932) and RF–ARIMA (MSE = 2.907, R^2 = 0.922). The comparison with deep learning approaches shows that long short term memory (LSTM) achieves MSE = 9.649 and R^2 = 0.879, followed closely by convolutional neural network (CNN)-LSTM (MSE = 13.224, R^2 = 0.834). In contrast, gated recurrent unit (GRU) (R^2 = 0.781), Transformer (R^2 = 0.781), and Wavelet-ARIMA (R^2 = 0.676) exhibit higher errors and weaker correlations, highlighting the relative advantage of temporal deep learning models for dynamic water quality forecasting. These results indicate that TSRF can improve the reliability and efficiency of WQI prediction, enabling policymakers and environmental managers to make informed, data-driven decisions and implement targeted pollution control strategies. Thus, this study can assist us to have better water quality management strategies by increasing the reliability and efficiency of WQI prediction. Water quality assessment methods developed in this research can help policymakers and environmental stewards to make informed decisions and implement targeted pollution control strategies. Advancing machine learning applications in environmental science strengthens current practices in water quality management and enhances decision-making capabilities.

Keywords Machine learning · Environmental monitoring · Predictive modelling · Forecasting and classification

Introduction

Water, which is one of the important resources on earth, is under a serious threat because of the cumulative effects of human activities and climate change (Zhang et al. 2022a, b). It is impossible to overstate the importance of water quality management on a worldwide scale. Water quality management is essential to the achievement of sustainable development, the protection of the environment, and the maintenance of public health. According to the United Nations (UN), the management of water quality is important to ensure the availability of water and sanitation for all people, as well as the management of these resources in a

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sustainable manner (United Nation 2023). The report by UN emphasises on the importance of achieving sustainable and equitable water resources management, specifically Goal 6 of the United Nations' 2030 Agenda for Sustainable Development (United Nation 2023). Among the measures highlighted are; better management of water resources and factoring water quality parameter values into decision-making. In fast developing countries like Malaysia, where industrial growth, urbanisation, and agriculture impose enormous demand on water resources, the issues of water quality management are particularly important.

Malaysia has tropical climate and high rainfall due to the geographical location and the country also faces rapid urbanization. Thus, water quality monitoring is significant and crucial in the country. The urbanization and weather condition lead to runoff, deforestation, and pollution from industrial and agricultural activities, impacting water resources. According to Abdul Maulud et al. (2021), it is essential to implement water quality management techniques to ensure undisrupted water supply, continuation of agriculture and industrial activities, and the creation of hydroelectric power.

In recent years, machine learning, a subfield of artificial intelligence (AI), has emerged as a powerful tool in environmental monitoring. It has opened new paths for the analysis of complicated environmental data. The ability of machine learning algorithm to process big datasets, learn from data patterns, and make prediction enables them to provide a dynamic and predictive approach for environmental monitoring. Machine learning models have shown their promise in water quality management by accurately predicting water quality parameters, detecting pollution sources, and even projecting future water quality scenarios (Khoi et al. 2022).

Traditional approaches to determine Water Quality Index (WQI) frequently depend on static parameters and predetermined criteria. Although these methods are advantageous for preliminary evaluations, they are deficient in the dynamic characteristics necessary for real-time or predictive monitoring (Jha et al. 2020). A primary disadvantage of conventional methods is their incapacity to manage extensive datasets or non-linear interactions effectively (Kadam et al. 2019). Furthermore, conventional WQI estimation is limited by static permitted thresholds that fail to consider variable environmental conditions (Kouadri et al. 2021). Utilising machine learning techniques, as outlined in this study, can address these constraints by managing intricate, multi-dimensional datasets and providing enhanced, real-time predictions (Chen et al. 2020).

Many researchers have studied the application of machine learning to forecast water quality previously. A recent review emphasizes that regression, classification, hybrid, and ensemble models often combined with techniques such as PCA and residual analysis have been widely applied for surface water quality prediction, particularly within Malaysia's environmental context (Lokman et al., 2025a). The study highlights that ensemble and hybrid approaches achieve superior accuracy, though challenges remain in data quality, interpretability, and the integration of spatio-temporal and fuzzy logic methods (Lokman et al., 2025a).

A potential way to improve the accuracy and effectiveness of water quality monitoring system is to incorporate these cutting-edge computational algorithms. The reliability and precision with ensemble approaches like Extra Trees Regressor and Random Forest (RF) can forecast environmental variables (Khoi et al. 2022). The research proves that Random Forest is good in dealing with complicated, biassed, and non-linear datasets. It makes the model ideal for multi-dimensional, non-linear water quality datasets. Kouadri et al. (2021) stated that Extra Trees is efficient in environmental data modelling, pointing out that it can effectively process big datasets and produce findings that are easy to understand.

One popular method for analysing time series data is the autoregressive integrated moving average (ARIMA) model. The term "time series" describes a set of collected data points that have been organised chronologically. For example, Wu et al. (2023) presented an ARIMA and clustering model for the purpose of river basin water quality prediction. On the other hand, Lokman et al. (2023) used the ARIMA modelling technique to help with smart toilet predictive maintenance, and then its efficacy was compared with the long short-term memory (LSTM) model. There is very little prediction error, and the model fits the data well.

A lot of people are interested in Artificial Neural Network (ANN) since the method is good in representing data with complicated, non-linear relationships. Long short-term memory (LSTM) network is a recurrent neural network (RNN), a machine learning algorithm that has been applied for water quality prediction. Khullar and Singh (2022) presented deep learning-based Bi-LSTM, a deep learning model constructed on a Bi-LSTM architecture to forecast the Yamuna River's water quality parameters. Malek et al. (2022) has evaluated several machine learning algorithms in predicting and categorising the water in the Malaysian Kelantan River Basin The algorithms used consisted of K-Nearest Neighbours (KNN), Support Vector Machine (SVM), Artificial Neural Network (ANN), Decision Tree (DT), Naïve Bayes (NB), Random Forest (RF), and Gradient Boosting (GB). Then, chemical oxygen demand (COD), biological oxygen demand (BOD), and ammoniacal nitrogen (NH3N) were the water quality parameters that had been studied. Four separate strategies are utilized to forecast the WQI such as M5P tree model (M5P), additive regression (AR), support vector machine (SVM), and random subspace

(RSS). The findings show that GB model is better compared to other models.

A new approach of water quality prediction using hybrid models, which integrates various machine learning strategies is proposed in Wu and Wang (2022). The author put forth a hybrid model that combines an artificial neural network with long short-term memory (ANN-LSTM) and a convolutional neural network with long short-term memory (CNN-LSTM) to forecast water quality parameters including total phosphorus, ammonia nitrogen, permanganate index, and dissolved oxygen. The results produced by the hybrid models outperform those of single models. A combined approach utilising decision trees and ARIMA was suggested by Patel et al. (2023) for the purpose of predicting water quality. This hybrid model, among many others, assessed water quality to obtain an improved water management system. Meanwhile, Random Forest Classifier is reported to outperform other machine learning models for predicting the quality of aquaculture water (Sen et al. 2023). The study also highlights the significance of tailoring model selection to individual use cases and data properties.

The models evaluated in this study are categorized into three main groups based on their methodological foundations:

- Traditional Regression Models:
  - *RR* was selected for its effectiveness in handling multicollinearity, which is often present in environmental datasets where water quality parameters may be highly correlated (Mohd Zebaral Hoque et al. 2022).
  - *ARIMA* is included due to its capability in capturing temporal trends and seasonality in univariate time series data, making it suitable for forecasting time-dependent water quality indices Wu et al. (2023).
- Machine Learning Models:
  - *DT* was chosen for its simplicity, transparency, and ability to model non-linear relationships between variables (Jeihouni et al. 2020).
  - *RF* is known for its ensemble learning mechanism that improves accuracy and robustness, particularly in noisy and high-dimensional datasets (Uddin et al. 2022).
  - *ETR* offers increased randomness during tree construction, which helps prevent overfitting and improves computational efficiency in large datasets (Khoi et al. 2022).
  - *ANN* was selected to explore the capability of deep learning approaches in modelling complex; non-linear relationships commonly found in water quality data (Wu and Wang 2022).
- Deep Learning Models:
  - Transformer was selected for its ability to model long-range dependencies in sequential data, improving prediction accuracy in complex aquatic environments (Li et al. 2025).
  - LSTM is effective in learning temporal patterns from time series water quality data, offering superior performance in dynamic trend modelling (Bu et al. 2025).
  - CNN was chosen for its spatial feature extraction capabilities and is often used in hybrid models for water quality forecasting across regions (Kow et al. 2025).
- Hybrid Models:
  - *DT-ARIMA* and *RF-ARIMA* integrate time series forecasting with decision-based models to capture both temporal dynamics and non-linear interactions, thereby enhancing predictive performance over individual models Patel et al. (2023).
  - The proposed *Time Series Regression Forest (TSRF)* combines Decision Tree, ARIMA, and Random Forest into a unified framework. This hybrid structure addresses both static and dynamic components of water quality data, offering improved generalization and accuracy. The design of TSRF is inspired by recent studies that demonstrate the effectiveness of combining statistical and machine learning models for forecasting environmental data (Zhou et al. 2022; Bai et al. 2024).

In comparison to traditional water quality prediction methods, machine learning models continue to demonstrate superior accuracy, efficiency, and adaptability in handling complex datasets. Traditional models, such as those relying on statistical or process-based techniques, often struggle with scalability and non-linear relationships. In contrast, machine learning models, including hybrid and deep learning approaches, have shown excellent results in processing multidimensional data, making them highly effective for forecasting water quality. For example, a recent study has integrated advanced machine learning models with explainable AI (XAI) techniques, significantly enhancing water quality assessment and prediction (Makumbura et al. 2024). It shows that the XGBoost model achieved $R^2 = 0.992$, mean absolute error (MAE) = 0.825, and root mean squared error (RMSE) = 1.381, outperforming conventional

weighted arithmetic water quality index methods. Another research effort demonstrated the use of blockchain technology alongside machine learning to provide secure and accurate water quality predictions (Geetha Jenifel 2024). It combined random forest with blockchain security, achieving MAE = 0.56, MSE = 0.33, and RMSE = 0.56, surpassing other tested models such as linear regression, linear discriminant analysis and support vector machine. Moreover, hybrid models combining traditional and deep learning methods proved particularly effective in predicting water quality in river systems with multiple surface parameters, outperforming conventional techniques due to their capability to generalize across spatial and temporal dependencies (Wang et al. 2024). It shows improving R^2 by ~29% and reducing RMSE by ~48.6% compared to traditional ML models, and further lowering RMSE by 18.1% relative to a single LSTM. These findings highlight the advantages of machine learning models, particularly their ability to handle complex data structures and provide dynamic, real-time insights for water quality management. Table 1 shows various previous

Table 1 Summary of previous studies applying machine learning techniques for water quality management, including the advantages and disadvantages of each approach and dataset size

References	Modelling techniques	Advantages	Disadvantages	Data size
(Fang et al. 2025)	LSTM	High accuracy in predicting water quality indicators across multiple basins.	Model tuning details are limited, reducing reproducibility.	Dataset size isn't stated but covers three rivers
(Abushandi 2025)	Artificial Neural Networks (ANN)	Employs a novel absorption characteristics recognition algorithm, delivering optimal regression performance.	The ACR method lacks detailed performance benchmarks regarding computational cost and robustness	200 spectral readings.
(Feng et al. 2025)	AI-driven Time Series (SVM & ARIMA) with Satellite Imagery	The hybrid framework combining Sentinel-2 imagery with SVM and ARIMA models increased spatial data coverage.	ARIMA is less optimal for non-seasonal parameters; limited subsurface insight	Dataset size isn't stated.
(Chen et al. 2024)	Deep Learning (LTSF-Linear Model)	Linear model baseline outperformed ARIMA, LSTM, and Informer, reducing MSE and MAE by 8.55% and 10.51%.	Limited to three parameters, restricting applicability to more complex multi-parameter models.	Hourly measurements from Jan 2022 to Jul 2023
(Bentley et al. 2022)	Proposed mass balance model to monitor the water quality at the river.	Provides a robust benchmark with 18,000 scenarios across varied network types.	Limited to chlorine dynamics; not evaluated for other water quality parameters.	18,000 simulation cases
(Haghnazar et al. 2022)	Case study on the effect of lockdown period on surface water pollution and quality.	Uses advanced AI (deep learning, ensemble) to directly predict WQI & water quality classes.	Details on data volume and parameter list are not fully disclosed	33,612 samples
(Ma et al. 2020)	Predicted the BOD using data from New York Harbor waters.	RMSE was 11.5–17.2% lower than regular matrix-completion approaches and 19.2–25.2% lower than classic machine learning models.	Deep models require extensive data to perform well	32,323 samples
(Zounemat-Kermani et al. 2019)	Predicted DO concentration using different water quality parameters.	Best hybrid reduces RMSE by 80% from standalone MLP	A small dataset size may limit broad applicability.	232 samples
(Pu et al. 2019)	Proposed water quality classification using Landsat8 images by CNN.	CNN provided superior classification over traditional ML approaches.	Temporal mismatch between satellite capture and sampling, plus weather variability.	481 samples
(Kumar et al. 2018)	Proposed the assessment of heavy-metal pollution in water to detect Pb, Cu, and Zn.	CNNs showed strong potential in mapping chlorophyll-a concentration from satellite data.	Deep learning models perform best with large, labeled datasets.	Dataset size isn't stated.

studies on the application of machine learning techniques for water quality management. All techniques are reviewed in terms of advantages and disadvantages and data size that have been studied.

From the literatures reviewed, there has been tremendous success in predicting water quality using machine learning methods. On the other hand, each model has strengths and weaknesses. Thus, the critical issues comprise of developing more efficient models computationally to ensure they can handle non-linear relationships and interpretable. This research expands upon the wealth of information by using a wide variety of machine learning models. It can provide reliable, effective, and practically useful recommendations for enhancing water quality index prediction and eventually improving the Malaysia's water quality management.

The aims of the research are given as follows:

1. To construct a novel model named as TSRF (Time Series Regression Forest) for the prediction of WQI in Malaysia. TSRF combines two regression models (decision tree and ARIMA) and one classification model (Random Forest).
2. To provide a solution to water quality monitoring that is more dynamic, accurate, and predictive for the purpose of overcoming the limits of traditional methods in predicting WQI.
3. To make a significant contribution for environmental science by designing and implementing an innovative approach that integrates conventional methods of evaluating water quality with cutting-edge machine learning strategies.

The anticipated contribution from the proposed model is significant. The finding can revolutionise water quality monitoring in Malaysia by utilising the predictive potential of machine learning. TSRF is employed to record both the continuous and categorical aspects of water safety levels, as well as the continuous nature of water quality assessments. There are different machine learning models that are used to compare the performance of TSRF model. Data gathering and preparation, machine learning techniques, model assessment, outcomes, and discussion are covered in the following sections, outlining the process of developing the TSRF model.

The organisation of this paper is as follows: The research methodology is presented in the next section, followed by the presentation and discussion of the outcomes. The following section discusses direction for future research and obstacles, and the summary is also provided in the final section.

Methodology

(a) Sources of water quality data.

Data from five critical areas of Klang, Semenyih, Labu, Muar, and Malacca River was collected from the Department of Environment (DOE), Malaysia. Additional data was retrieved at the MRANTI Lake location. The comprehensive location map is depicted in Fig. 1.

The places are selected for their diverse dataset, which is made possible by their varying degrees of urbanisation and industrialization. The water quality in Klang is affected by pollutants from ships and factories, and the city's high population density makes its condition even worse. Although Semenyih has experienced urbanisation, it is primarily recognised for its agricultural pursuits including both farm discharges and urban runoff. Labu sheds light on the effects of semi-rural water quality on areas with lower population densities and mixed-use land (agricultural and light industry). Muar is well-known for its furniture industry and historical value, but its water quality statistics shows that the pollution from cities and factories has affected the area. Urban runoff and tourism-related activities impact the Malacca River as it flows through historically rich urban areas. Although the MRANTI lake region serves as a technology and tourism park, it is nonetheless vulnerable to urban runoff and trash and probably contributes less directly to industrial pollution.

Six surface water quality parameters are included in the system, which are pH, dissolved oxygen, total dissolved solids, ammoniacal nitrogen, biochemical oxygen demand, and chemical oxygen demand. Table 2 shows the parameters range according to the Malaysia's National Water Quality Standard for water classification.

(b) Cleaning, Normalization and Handling Missing Values and Outliers.

Data collected were subjected to a rigorous cleaning process to ensure accuracy and consistency. This involved correcting discrepancies, aligning units of measurement, and standardizing formats. The data were then normalized using the Min–Max normalization method, scaling each parameter to a range of 0 to 1. This ensured that all features contributed equally to the analysis, preventing parameters with larger numerical ranges from dominating the model training. Missing values were addressed using mean imputation for continuous parameters and mode imputation for categorical parameters, depending on the nature of the variable.

Z-Score serves as a normalization method that relies on the mean and standard deviation of the data. It effectively reduces the impact of distribution outliers from the



Fig. 1 Map showing the six locations of water quality data collection

Table 2 Water quality index classification based on malaysia's National water quality standard (DOE 2020; Kamarudin et al. 2020)

PARAMETER	UNIT	CLASS				
		I	II	III	IV	V
Ammoniacal Nitrogen	mg/l	<0.1	0.1–0.3	0.3–0.9	0.9–2.7	>2.7
Biochemical Oxygen Demand	mg/l	<1	1–3	3–6	6–12	>12
Chemical Oxygen Demand	mg/l	<10	10–25	25–50	50–100	>100
Dissolved Oxygen	mg/l	>7	5–7	3–5	1–3	<1
pH	-	>7.0	6.0–7.0	5.0–6.0	<5.0	>5.0
Total Suspended Solid	mg/l	<25	25–50	50–150	150–300	>300
Water Quality Index (WQI)		>92.7	76.5–92.7	51.9–76.5	31.0–51.9	<31.0

transformation results and proves to be very useful when the minimum and maximum actual values of the data are unknown, as demonstrated in Eq. (1) where $\bar{x}$ is a mean value observed, μ is a population mean and σ is a standard deviation (Pagan et al. 2023). Those anomalies are removed to ensure that extreme values do not skew model performance. Lagging variables are created for water parameters to capture temporal dependencies. The variables enable the model to consider historical data points during predictions (Lin et al. 2024).

$$z = \frac{\bar{x} - \mu}{\sigma} \tag{1}$$

(c) Water Quality Classification (WQC).

The Water Quality Classification (WQC) was established by DOE, Malaysia, using the rating determined by the WQI. The focus of the current analysis was on the WQC, which was considered the dependent variable. The quality of water can be evaluated by utilising the WQI. According to Table 2, water classified as class I is characterised by its high cleanliness, with minimal levels of pollutants across all water parameters. The water classified as class II remains pure, albeit with a slightly elevated concentration of pollutants. The product is deemed appropriate for engaging in activities such as swimming and is deemed suitable for human utilisation. Water classified as class III necessitates a more comprehensive treatment process prior to its utilisation. This water is appropriate for species with lower sensitivity and for agricultural applications. Class IV water exhibits substandard quality and necessitates sophisticated treatment. The levels of pollutants in this water are detrimental to delicate aquatic organisms and unsuitable for human consumption. Water classified as class V exhibits the

most severe water quality, characterised by elevated levels of contaminants.

(d) Water Quality Index (WQI) calculation.

In this research, the WQI is calculated following Malaysia Water Quality Standard (DOE 2020) since the data was collected in Malaysia. The WQI formula to comprehensively assesses water quality as provided by the DOE, Malaysia (Chia et al. 2022; DOE 2020) is given in Eq. (2).

$$WQI = (0.22 * SI_{DO}) + (0.19 * SI_{BOD}) + (0.16 * SI_{COD}) + (0.15 * SI_{AN}) + (0.16 * SI_{SS}) + (0.12 * SI_{pH}) \quad (2)$$

The sub-index (SI) values of water quality characteristics are combined in the WQI formula. This process yields a single numerical value that represents the overall water quality. The weight of these criteria is established according to the DOE Standard. WQI standards vary among countries due to variations in local environmental conditions, industrial activities, and public health goals. While several countries adhere to the criteria set by the World Health Organisation, particular countries such as the United States or Malaysia may establish their own distinct standards to solve local water quality issues (DOE 2020). SI_{DO} refers to Subindex Dissolved Oxygen, SI_{BOD} is Subindex Biochemical Oxygen Demand, SI_{COD} is Subindex Chemical Oxygen Demand, SI_{AN} is Subindex Ammoniacal Nitrogen, SI_{SS} is Subindex Suspended Solid, and SI_{pH} is Subindex pH. These sub-indices are obtained using Eq. (3) to Eq. (8) that determine the best-fit connection for each water quality parameter. Each water quality parameter is weighted differently to determine the WQI, indicating its relative importance to the overall water quality evaluation for certain users, such as drinking water, recreation, or aquatic life. The accuracy of WQI to water quality parameter changes depends on the weighting mechanism.

$$SI_{DO} = \left\{ \begin{array}{ll} 0 & for\ D0 \leq 8 \\ 100 & for\ D0 \leq 92 \\ -0.395 + 0.0.0D0^2 - 0.00020D0^3 & for\ 8 < D0 < 92 \end{array} \right\} \quad (3)$$

$$SI_{BOD} = \left\{ \begin{array}{ll} 100.4 - 4.23BOD & for\ BOD \leq 5 \\ 108e^{-0.055BOD} - 0.1BOD & for\ BOD > 5 \end{array} \right\} \quad (4)$$

$$SI_{COD} = \left\{ \begin{array}{ll} -1.33COD + 99.1 & for\ COD \leq 20 \\ 103e^{-0.0157COD} - 0.04COD & for\ COD > 20 \end{array} \right\} \quad (5)$$

$$SI_{AN} = \left\{ \begin{array}{ll} 100.5 - 105AN & for\ AN \leq 0.3 \\ 94e^{-0.573AN} - 5\,|AN - 2| & for\ 0.3 < AN < 4 \\ 0 & for\ AN \geq 4 \end{array} \right\} \quad (6)$$

$$SI_{SS} = \left\{ \begin{array}{ll} 97.5e^{-0.00676SS} + 0.05SS & for\ SS \leq 100 \\ 71e^{-0.0016SS} - 0.015SS & for\ 100 < SS < 1000 \\ 0 & for\ SS \geq 1000 \end{array} \right\} \quad (7)$$

$$SI_{pH} = \left\{ \begin{array}{ll} 17.2 - 17.2pH + 5.02pH^2 & for\ pH < 5.5 \\ -242 + 95.5pH - 6.67pH^2 & for\ 5.5 \leq pH < 7 \\ -181 + 82.4pH - 6.05pH^2 & for\ 7 \leq pH < 87.5 \\ 536 - 77.0pH + 2.76pH^2 & for\ pH \geq 8.75 \end{array} \right\} \quad (8)$$

(e) Machine Learning Models.

The structured classification and literature-backed rationale provide a clear basis for the inclusion of each model and support the comparative evaluation presented in subsequent sections. Our dataset includes 11,065 samples with different water quality parameters such as pH value, dissolved oxygen, chemical oxygen demand, biochemical oxygen demand, ammoniacal nitrogen and total dissolved solid. All analyses were performed in Python 3.10 using the following libraries: pandas (v1.5.3) for data handling, numpy (v1.24.2) for numerical computations, scikit-learn (v1.2.2) for implementing the Random Forest model, statsmodels (v0.13.5) for the ARIMA model, and matplotlib (v3.7.1) and seaborn (v0.12.2) for data visualisation. Hyperparameter tuning for Random Forest was conducted using the GridSearchCV function from scikit-learn. The overall system is shown in Fig. 2 which outlines the key components and their relationships in our approach.

The study employs a diverse set of machine learning models to predict water quality parameters effectively:

1. The Extra Trees Regressor (ETR) operates by constructing a multitude of randomized decision trees. At each split in the decision tree, the model selects the best feature among a random subset of features, rather than searching for the most significant feature while splitting a node (Khoi et al. 2022). This randomness helps in increasing the model's robustness and reduces the variance, making it less likely to overfit on the training data. In the context of WQI prediction, the Extra Trees Regressor can capture complex, nonlinear relationships between various water quality parameters and the WQI, providing a comprehensive model that accounts for the randomness and variability, inherent in environmental data.
2. Ridge Regression (RR) addresses some of the common problems in regression, particularly multicollinearity. By adding a degree of bias to the regression estimates, Ridge Regression minimizes the complexity of the model and prevents overfitting, which is crucial for a predictive model (Mohd Zebaral Hoque et al. 2022). It introduces a penalty term (alpha) to the loss function, forcing the model to keep the model weights as small as possible. In predicting WQI, Ridge Regression can provide stable and interpretable results, especially when dealing with data where predictor variables are highly correlated.

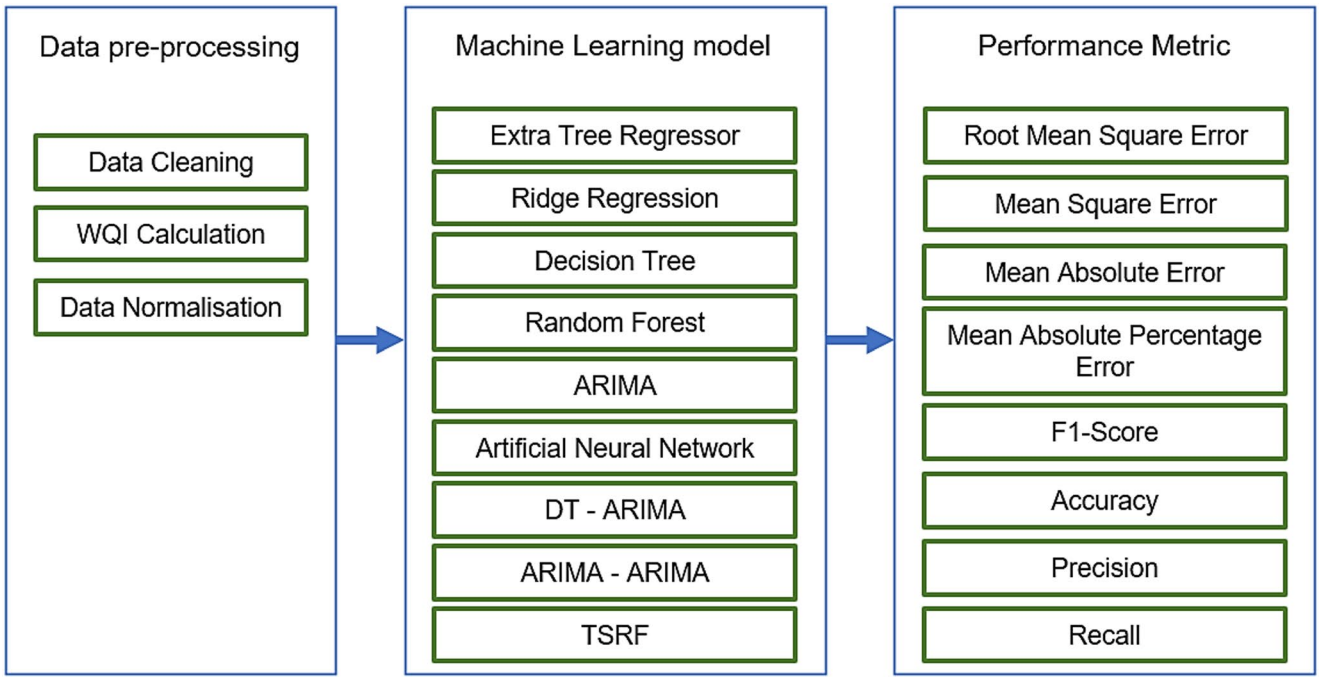


Fig. 2 Flowchart of the developed method

3. Random Forest (RF) is an ensemble learning technique, building multiple decision trees and merging their outputs to improve the overall result. The model works on the principle of bagging, which helps in reducing variance and avoiding overfitting (Uddin et al. 2022). Each tree in the Random Forest gives a class prediction or mean prediction, and the class or mean with the most votes becomes the model's prediction. For WQI prediction, the Random Forest model can handle a mix of categorical and continuous variables and automatically handle missing values, making it a flexible and powerful option for complex environmental datasets.
4. ANNs are inspired by the biological neural networks in the human brain, consisting of input layers, hidden layers, and output layers. Each node in a layer is connected to nodes in the previous and next layer, and these connections have associated weights. During training, the network adjusts these weights based on the error of the output compared to the expected result. ANNs are highly flexible and can model complex nonlinear relationships. They are particularly useful in water quality prediction when there are intricate patterns and relationships between parameters that affect the WQI.
5. ARIMA, a short form for Auto-Regressive (AR) Integrated Moving Average (MA), is a forecasting algorithm that is based on the idea that information in the past values of the time series can alone be used to predict future values. ARIMA models are characterized by three primary parameters: p (AR terms), d (differencing order), and q (MA terms). This model is well-suited for univariate time series data that show patterns and is not stationary, where data values depend on their previous values. In the context of WQI prediction, ARIMA can be particularly effective if the index shows clear trend or seasonal patterns over time (Ghaemi et al. 2022). Equation (9) shows the overall equation of ARIMA model (Ospina et al. 2023).

$$Y_t = c + \emptyset_1 Y_{t-1} + \ldots + \emptyset_p Y_{t-p} + \theta_{1}\, {}_{t-1} + \ldots + \theta_{q}\, {}_{t-q} + \quad {}_{t} \tag{9}$$

where Y_t is the observation at time t, $\varnothing_j$ is the parameter of the autoregressive part of the model, θ_k is the parameter of the moving average part, and ϵ_t is the error term.

6. A DT is a flowchart-like tree structure where each internal node represents a "test" on an attribute, each branch represents the outcome of the test, and each leaf node represents a class label (decision taken after computing all attributes). The paths from the root to the leaf represent classification rules. In predicting WQI, a Decision Tree model will learn to infer the conditions that lead to specific water quality outcomes, making it a transparent and easy-to-interpret model for decision-making (Jeihouni et al. 2020).

7. Hybrid models like RF-ARIMA and DT-ARIMA combine the strength of machine learning models (Random Forest or Decision Tree) and time series forecasting models (ARIMA) to make prediction. The machine learning component can capture the complex relationships between different water quality parameters, while the ARIMA component can model the time-dependent trends in the WQI. These hybrid models are particularly powerful to capture both the underlying trend and the complex interaction in your dataset, offering a comprehensive approach to predict WQI. The ANN, Ridge Regression, and Extra Trees Regressor are not suitable for a hybrid model due to their high capacity and tendency to overfit.

(f) Proposed model, Time Series Regression Forest (TSRF).

TSRF is unique where it takes a domain-specific approach to water quality index monitoring regression and classification problems. The TSRF model, which combines decision tree (DT), ARIMA, and random forest (RF), is validated by extensive prior research highlighting the efficacy of these techniques in predicting water quality. Recent research has shown the efficacy of these models in environmental forecasting. A study has established a hybrid prediction framework for water quality by combining wavelet decomposition, ARIMA, and gated recurrent unit (GRU) models, resulting in enhanced forecast accuracy and stability relative to conventional models (Zhou et al. 2022). Furthermore, a study on stream water quality prediction which utilises boosted regression trees and RF models has demonstrated encouraging outcomes, highlighting their resilience and versatility in managing intricate environmental data (Alnahit et al. 2022). Then, a study which evaluates the accuracy of water quality prediction using RF regression and Artificial Neural Networks, has found that the RF model has a lower mean squared error, signifying enhanced performance (Hai et al. 2023). A recent study has introduced a linked prediction model utilising RF feature selection for river water quality forecasting, demonstrating the model's efficacy in managing the complexities and nonlinear variations in water quality predictions (Bai et al. 2024). The use of RF and DT in the TSRF model is substantiated by their demonstrated efficacy in handling heterogeneous environmental data and producing meaningful insights for water quality management.

The overall flowchart is shown in Fig. 3. The procedure commences with the collection of data for the purpose of monitoring water quality. These variables are essential for maintaining the health and safety of water bodies. Crucial preprocessing measures involve the removal of irrelevant information and the handling of missing data following the acquisition of data. Analytical models are limited to handling only datasets that consist of numerical values. Entries that are not numerical are transformed into numerical values. The WQI is determined based on the provided data. The WQI provides a comprehensive evaluation of water quality by consolidating multiple measurements of water parameters into a single result.

The training phase of the predictive models utilises 70% of the dataset for model fitting, while the remaining 30% is allocated for testing and validation. To avoid temporal leakage, the split was performed chronologically, with earlier records used for training and later records used for testing. Equation (10) shows the proposed ARIMA model followed by overall TSRF model in Eq. (11).

$$\hat{y}_t^{ARIMA} = c + \sum_{i=1}^{p} \emptyset_i \hat{y}_{t-1}^{DT} + \sum_{j=1}^{p} \emptyset_j \varepsilon_{t-j} \tag{10}$$

$$\hat{y}^{TSRF} = majority\ vote \left(h_1\left(\hat{y}_t^{ARIMA}\right), h_2\left(\hat{y}_t^{ARIMA}\right), \ldots, h_m\left(\hat{y}_t^{ARIMA}\right)\right) \tag{11}$$

where $\hat{y}_t^{ARIMA}$ is the ARIMA-predicted WQI at time t. $\hat{y}_{t-1}^{DT}$ is the WQI predicted by the Decision Tree at lag $t-1$. $\varnothing_i$ is autoregressive coefficient and $\varnothing_j$ is moving average coefficient. $h_1, h_2, \ldots, h_m$ are individual decision trees. Random Forest uses majority voting to classify the final WQI class based on $\hat{y}_t^{ARIMA}$. In this architecture, ARIMA uses lagged DT predictions rather than the raw WQI series. This design is chosen because DT captures complex, nonlinear relationships between WQI and its input parameters, producing residual patterns with clearer temporal structure. Then, ARIMA models these temporal dependencies more effectively than when applied directly to the raw series, improving forecast stability and reducing noise.

The training data is used as input into a Decision Tree model, which is then used to predict or estimate water quality. Decision Trees in supervised learning are used to model decision rules based on data. Data scaling is a technique used to standardise feature values, making them comparable and optimising various modelling processes after training the model. The time series characteristics of the DT predictions were examined using Autocorrelation (ACF) and Partial Autocorrelation (PACF) plots. The ACF and PACF plots (Figs. 4 and 5) are used to guide the ARIMA parameter selection. The ACF plot shows significant autocorrelation values up to lag 4 before dropping within the 95% confidence bounds, suggesting that temporal dependencies are captured primarily within the first four lags. The PACF plot

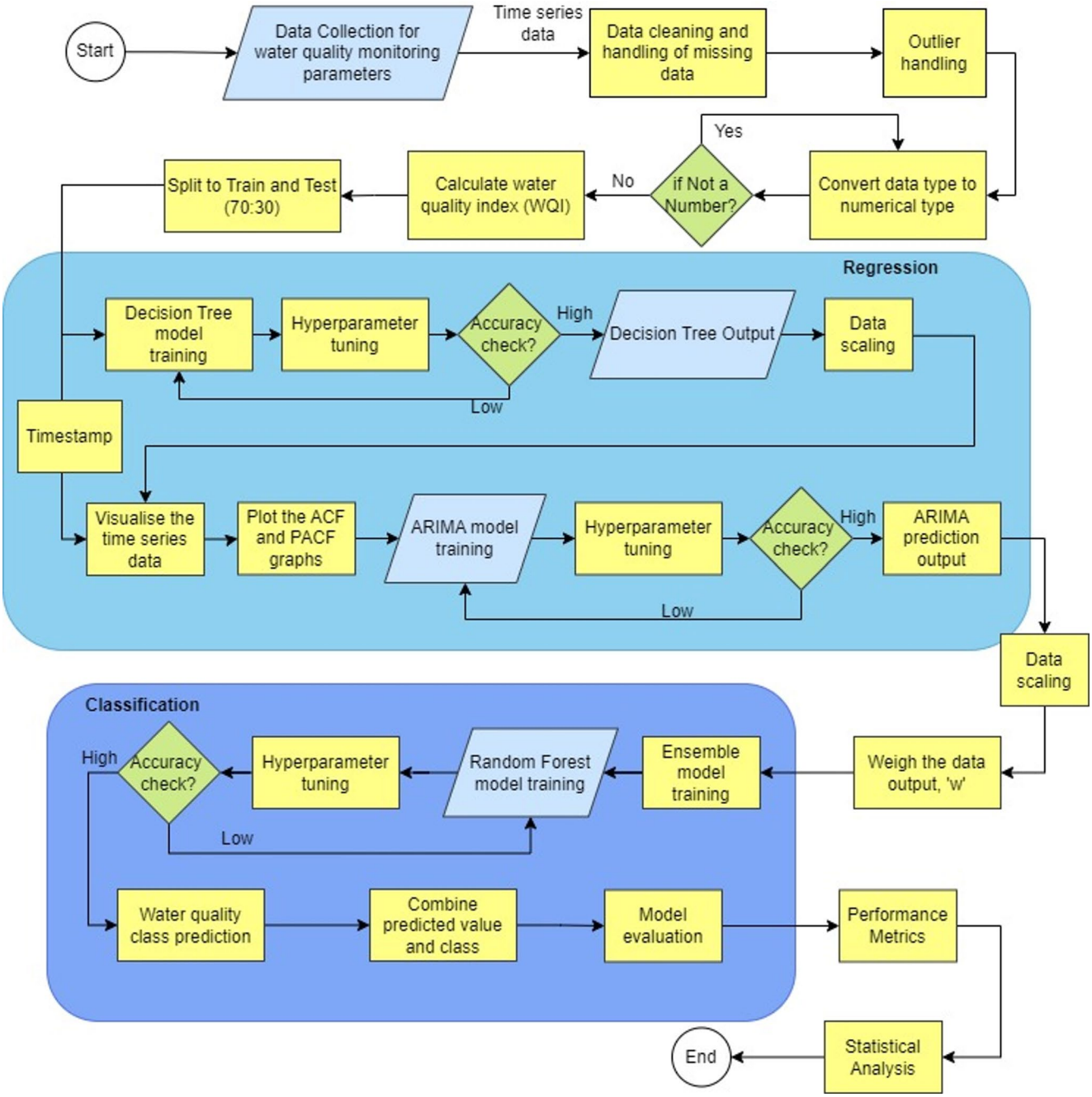


Fig. 3 The flowchart of TSRF model

exhibits a sharp cut-off at lag 4, a characteristic indicator of the autoregressive order $p=4$. Additionally, the gradual decay pattern observed in the ACF supports the selection of the moving average order $q=3$. These observations confirm that the chosen ARIMA (4,0,3) configuration is appropriate for modelling the temporal patterns in the lagged Decision Tree predictions within the TSRF framework.

The ARIMA model utilises scaled and weighted output data to calibrate its predictions to align with the original data. Regression is replaced by classification in the study. The Random Forest algorithm, which is known for its high accuracy, is used to estimate water quality based on the available data. The Random Forest model outperforms a single Decision Tree by mitigating overfitting. The setting parameters are shown in Table 3. The Decision Tree Regression component is configured with a maximum depth of 10, which ensures the tree doesn't overfit while maintaining flexibility to capture non-linear relationships between the water quality parameters and WQI. The ARIMA model ($p=4$,$q=0$,$q=3$) is employed to handle the time-series

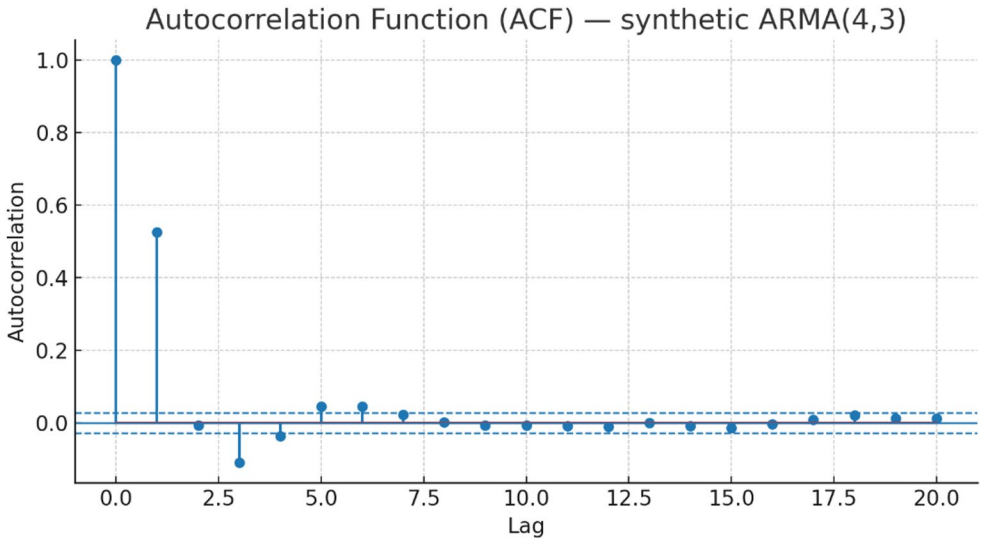


Fig. 4 ACF of the lagged DT predictions used as input to the ARIMA model. Significant autocorrelation is observed up to lag 4, with values exceeding the 95% confidence bounds

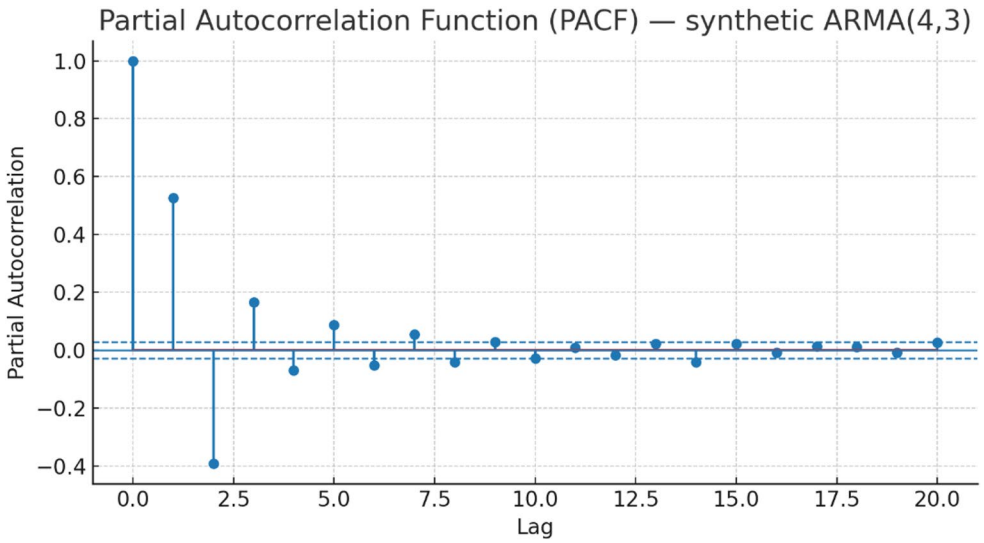


Fig. 5 PACF of the lagged DT predictions used as input to the ARIMA model. A clear cut-off is observed at lag 4, indicating the appropriateness of setting the AR order parameter $p=4$

Table 3 Setting parameters for TSRF

Model	Parameters
TSRF	DT: max Depth: 10; min samples split: 2; max features: "auto"
	ARIMA: Value of $p=4$, d=0, q=3,
	RF: n_estimators: 100; max depth: 10; min samples split: 2; classification output: 5 Classes (Class I – V)

aspect of the WQI prediction. This combination allows the model to capture temporal dependencies, improving the prediction of WQI based on historical data trends. Finally, the Random Forest Classification component is configured with 100 decision trees. It classifies the continuous WQI predictions into one of five water quality classes (Class I to V) based on the majority vote among the trees. Each tree in the forest has a maximum depth of 10 to prevent overfitting. This classification step ensures the predicted WQI is mapped to discrete water quality levels, enabling clear categorization of water safety levels.

The procedure encompasses precision verification and evaluation of the model. Model evaluation involves assessing the predictive ability of the model and its ability to generalise to new data. Performance metrics such as accuracy, precision, recall, and F1-score are utilised in this stage. A comprehensive statistical analysis is conducted on the model and its prediction. This analysis examines the statistical significance of the results to verify the reliability of the model's conclusions.

The goal of the TSRF regression is to forecast the WQI value using six important water parameters: DO, BOD, COD, pH, SS, and ammoniacal nitrogen. The advantages of TSRF in comparison to other models are as follows:

1. The temporal dynamics and autocorrelations of water quality data can be captured by TSRF through the application of ARIMA models. Non-stationarity and underlying cyclical or seasonal patterns can be easily identified

by ARIMA models (Singh et al. 2024). The model may adapt to more complicated and conditional data structures that may be time-dependent when coupled with a Decision Tree for regression, which handles nonlinear interactions between the parameters and the WQI.

2. Random Forest technique is employed in the classification portion of TSRF to classify the WQI class. The durability of Random Forests against overfitting caused by bagging and feature randomness, their capacity to run in parallel, and their excellent accuracy have made them well-known (Prati et al. 2025). Without being unduly affected by background noise, they can record intricate interplays between variables that can affect the water quality class.
3. By combining both models into TSRF, we can reap the benefits of both predictive modelling and time series forecasting. Previous values frequently impact future readings which are important in water quality monitoring. The hybrid model integrates both the individual relationships of the model and the temporal sequence of data.

The design of TSRF is unique since it addresses both the static and dynamic parts of water quality data, in contrast to models like ETR, RR, a single DT, or an ANN. Static parts of the data are characteristics that remain relatively consistent over time, such as baseline parameter levels. Dynamic parts reflect fluctuations or anomalies over time, capturing temporal dependencies and short-term variations (Yi et al. 2025).

In TSRF, the static part is handled by regression trees (DT) that model stable trends and steady-state behaviour, while the dynamic part is captured through ARIMA, which models temporal dependencies. To examine the role of each component, an ablation-style analysis was conducted in which the static and dynamic modules were evaluated separately and with combination. The performance comparison of these configurations is presented in the Section: Results and Discussion, allowing the relative contribution of the RF-based static component and the ARIMA-based dynamic component to be quantified. While ANNs can be complex and data-intensive to train, Decision Trees alone lack the ensemble robustness of RF, making them more vulnerable to overfitting. Table 4 presents the hyperparameters used for all baseline models to ensure consistent implementation and fair comparison, with settings determined from prior literature and light tuning via grid search.

Table 4 Hyperparameters of comparison models

Model	Key Hyperparameters
DT	max_depth = 10, min_samples_split = 2, max_features="auto"
RF	n_estimators = 100, max_depth = 10, min_samples_split = 2, max_features="auto"
ETR	n_estimators = 100, max_depth = 10, min_samples_split = 2, max_features="sqrt"
RR	alpha = 1.0, solver="auto"
ARIMA	p = 4, d = 0, q = 3 (same as in TSRF)
ANN	hidden_layers=(100, 50), activation="relu", solver="adam", alpha = 0.0001, max_iter = 200
DT-ARIMA	DT: same as above; ARIMA: p = 4, d = 0, q = 3
RF-ARIMA	RF: same as above; ARIMA: p = 4, d = 0, q = 3

(g) validation of regression method

For each model, the dataset is split into a training set and a testing set. The training set is used to train the model, and the testing set is used to evaluate its performance. Mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), mean square error (MSE) and correlation coefficient (R^2) are used to assess the accuracy of the data, with corresponding Eq. (12) to Eq. (16) (Wu and Wang 2022);

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|\hat{Y}_i - y_i| \tag{12}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{Y}_i - y_i)^2} \tag{13}$$

$$MAPE = \frac{1}{n}\frac{\sum_{i=1}^{n}\left|\hat{Y}_i - y_i\right|}{y_i} \tag{14}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(\hat{Y}_i - y_i)^2 \tag{15}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{Y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{Y}\right)^2} \tag{16}$$

In the equations, n represents the total number of data points, $\hat{Y}_i$ represents the model's projected value for that data point, y_i represents the actual data value, and $\bar{Y}$ represents the average data point. In addition to the standard 70% training and 30% testing split, we implemented a k-fold cross-validation procedure (k = 10) to provide a more robust estimate of the model's generalization performance. The dataset was randomly divided into 10 equal folds, with each fold used once as the validation set while the remaining folds served as the training set. The results from each fold were averaged to obtain cross-validated performance metrics. This approach reduces overfitting risk and ensures that the model's predictive performance is not overly dependent on a single train–test split.

To further examine the possibility of overfitting, we performed an analysis of the residuals in relation to the fitted values. A well-distributed pattern of residuals, appearing random and without any visible trend, indicates that the model is not capturing noise or overfitting the training data. This step was taken to complement the cross-validation process and ensure the reliability of the model's predictions.

(h) Validation *of classification method*.

The evaluation of classification model performance involves the utilisation of four distinct measures, namely accuracy, sensitivity/recall, precision, and F1-score. The metrics are expressed in the following equations, Eq. (17) until Eq. (20). (Islam Khan et al. 2022):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{17}$$

$$Sensitivity/Recall = \frac{TP}{TP + FN} \tag{18}$$

$$Precision = \frac{TP}{TP + FP} \tag{19}$$

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{20}$$

In this context, TP represents the number of true positive instances, TN represents the number of true negative instances, FP represents the number of false positive instances, and FN represents the number of false negative instances.

Results and discussion

(a) Analysis of data based on water quality parameters.

The dataset (11,065 samples in total) is sufficiently large to provide a strong foundation for models to forecast water quality. The dataset covers all important variables related to water quality. In this study, we find the standard deviation, minimum, maximum, and quartiles (Q1, Q2, Q3) for pH, total soluble solids (TSS), ammoniacal nitrogen (AN), biochemical oxygen demand (BOD), chemical oxygen demand (COD), and dissolved oxygen. All metrics are used for more in-depth analyses and studies into the water quality dynamics in the examined area by providing information on the mean, standard deviation, and distribution of each parameter. This is shown in Table 5.

Then, Fig. 6 shows the entire dataset using the histogram method. The histograms provide insights into the frequency distribution of each parameter:

- pH: Appears to be slightly left-skewed, indicating a higher frequency of values on the lower side of the pH scale. The water could be considered mildly acidic conditions which can affect the health and diversity of aquatic life.
- DO (Dissolved Oxygen): The distribution seems relatively uniform with a slight peak, suggesting a broad range of DO levels in the water samples. Most aquatic life cannot survive without enough DO. Stress and mortality among aquatic organisms caused by low DO levels can affect ecosystem structure and reduce biodiversity. Therefore, the WQI would be negatively affected, indicating poor water quality, in regions within the tested range when DO levels are low.
- BOD (Biological Oxygen Demand): Shows a right-skewed distribution, indicating that most water samples have lower BOD values, with fewer samples having higher BOD levels. More oxygen is accessible for aquatic creatures when the BOD levels are low. Ensuring the water body can support life, this condition supports a rich and resilient aquatic ecosystem, which positively contributes to the Water Quality Index (WQI).
- COD (Chemical Oxygen Demand): It is right-skewed, similar to BOD, indicating that while most samples have lower COD values, there are a few samples with significantly higher values. High COD levels indicate a potential for oxygen depletion as chemicals are broken down, which can harm aquatic life.
- TSS (Total Soluble Solids): It is normally distributed with a slight left skew. Since the distribution is slightly

Table 5 Descriptive data for water quality parameters

	DO (mg/l)	BOD (mg/l)	COD (mg/l)	TSS (mg/l)	pH	AN (mg/l)
Mean	6.8387	3.4139	31.1416	23.5796	6.5297	0.3787
Standard Deviation	1.2494	1.3323	12.6834	23.5710	0.8164	0.3654
Min	0.0000	0.5000	1.0000	0.0000	0.0000	0.0090
Q1	6.2700	2.6200	23.4500	18.1000	6.4000	0.27200
Q2	6.90000	3.340000	30.6900	24.0000	6.7000	0.3460
Q3	7.6000	4.0300	37.3900	26.0000	6.80000	0.4140
Max	9.9000	17.0000	110.0000	1280.0000	7.8700	10.7000

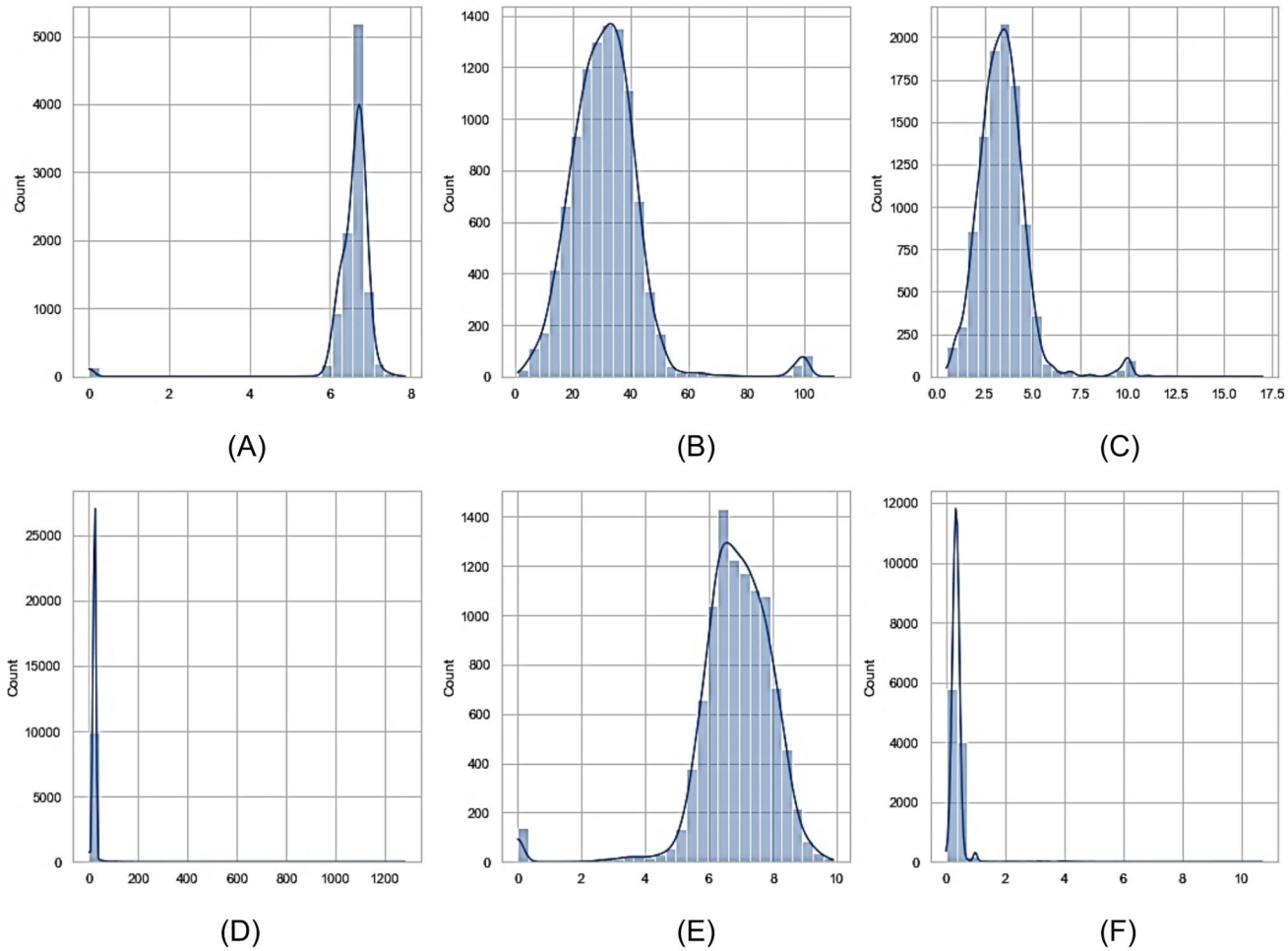


Fig. 6 Visualisation of the entire dataset using histogram method. **A** pH, **B** COD, **C** BOD, **D** TSS, **E** DO and **F** AN

left-skewed, it indicates generally favourable conditions for the WQI, if TSS levels remain within acceptable ranges for specific uses.

- AN (Ammoniacal Nitrogen): Appears to be normally distributed, centred around the mean. This stability is generally positive for the WQI, as it indicates the absence of excessive ammonia pollution.

Visible outliers were detected using the Z-score method, applying a threshold of ±3 standard deviations from the mean. These values were carefully examined and determined to represent genuine extreme environmental conditions, such as episodic pollution events or seasonal fluctuations, rather than data entry errors. For this reason, they were retained in the dataset to preserve the natural variability and capture rare but important water quality conditions. The study that applies statistical tests such as Shapiro–Wilk, Breusch–Pagan, PCA, and feature importance to address non-normality, heteroscedasticity, and dimensionality in water quality prediction (Lokman et al., 2025b).

The correlation matrix which visualizes the relationships between the parameters is shown in Fig. 7. The value of 1 or −1 indicate a strong positive or negative correlation, respectively. Then, the value closes to 0 indicates no linear correlation.

From the correlation matrix:

- DO and BOD: The parameters are negatively correlated (−0.52), as expected. Higher BOD depicts lower oxygen levels due to increased oxygen consumption by microorganisms.
- COD and BOD: The parameters show a positive correlation (0.66), suggesting that when the organic matter (as measured by BOD) is high, the COD is also high.

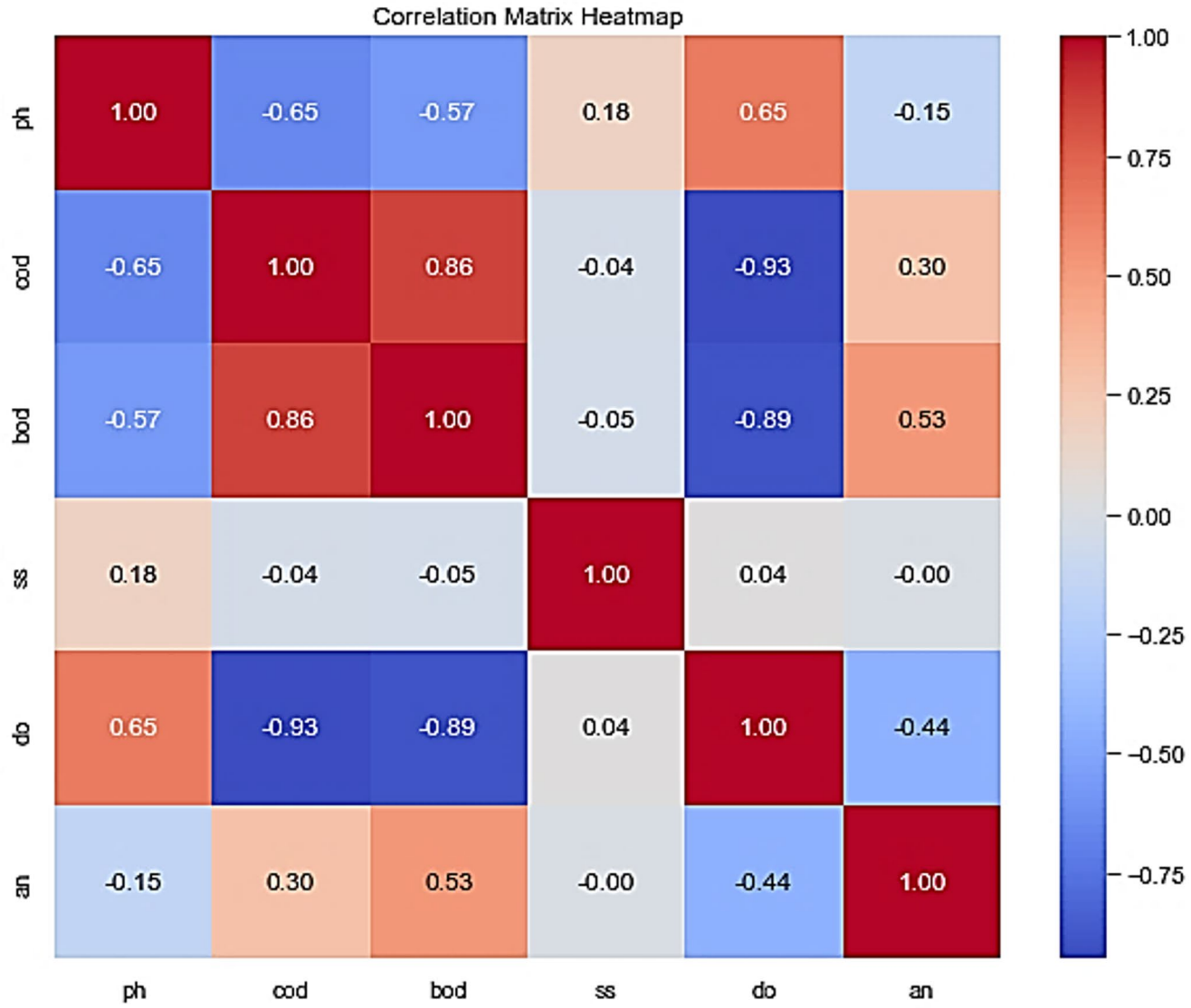


Fig. 7 Descriptive data for water quality parameters

Table 6 Feature importance of water quality parameters for WQI prediction using three methods: RF Gini impurity, permutation importance, and mutual information. Higher values indicate greater contribution to model predictions

Feature	RF Gini	Permutation	Mutual Information
COD	0.4977	0.8662	0.2123
BOD	0.2177	0.6011	0.1818
TSS	0.1509	0.7081	0.1834
pH	0.0776	0.0581	0.8324
DO	0.028	0.0206	0.3042
AN	0.028	0.9699	0.5248

- TDS: The parameter shows relatively low correlation, indicating that the number of dissolved solids does not strongly relate to the oxygen levels, organic matter, or pH.
- AN: The parameter shows relatively low to moderate correlation with other parameters.

While the correlation matrix (Fig. 7) summarizes linear relationships between water quality parameters, it does not fully capture their predictive contributions in non-linear models. To enhance model explainability, we computed feature importance scores using three complementary approaches: Random Forest Gini impurity, permutation importance, and mutual information. RF Gini quantifies the average contribution of each feature to reduce node impurity across all trees, while permutation importance measures the increase in prediction error when a feature's values are randomly shuffled. Mutual information estimates the amount of shared information between each feature and the target variable, capturing both linear and non-linear dependencies.

Table 7 Performance of machine learning models for regression

Model name	MSE $(mg/L)^2$	MAE (mg/L)	RMSE (mg/L)	MAPE (%)	R^2
Extra Tree Regressor (ETR)	12.1130	2.1770	3.4800	2.5250	0.5810
Ridge Regression (RR)	91.4540	2.3190	9.5630	2.9960	−2.1570
Random Forest (RF)	11.5650	1.9730	3.4000	2.3850	0.6010
ANN	19.4350	2.3150	4.4080	2.8670	0.3290
ARIMA	4.8300	1.1390	2.1970	1.4330	0.8710
Decision Tree (DT)	9.3300	1.4800	3.0550	1.8480	0.6770
DT-ARIMA	2.5260	0.3990	1.5890	0.4980	0.9320
RF-ARIMA	2.9070	0.6280	1.7050	0.7690	0.9220

Results (Table 6) show that COD and BOD have the highest predictive influence across all methods, with COD consistently ranking first (RF Gini: 0.4977; Permutation: 0.8662). TSS ranks third in RF Gini (0.1509) but is also important in permutation analysis, indicating its role in capturing variability unexplained by organic matter parameters. pH shows relatively low Gini importance (0.0776) but high mutual information (0.8324), suggesting non-linear effects on WQI. DO and AN have low Gini scores but AN ranks highest in permutation importance (0.9699), which may reflect context-specific interactions in the dataset. This multi-metric approach confirms that while COD and BOD dominate WQI prediction, other parameters contribute unique predictive signals.

(b) Analysis of various machine learning models.

Performance of various models in terms of MSE, MAE, RMSE and MAPE is shown in Table 7. According to Table 7, the Decision Tree - ARIMA (DT-ARIMA) and Random Forest - ARIMA (RF-ARIMA) models have the lowest MSE, MAE, RMSE, and MAPE. Compared to other models, the hybrid models are far more accurate and dependable for the provided dataset. DT-ARIMA and RF-ARIMA are consistent across all four metrics and have the lowest error metrics. It shows a strong indication of their robustness. The dataset may benefit from a combination of conventional machine learning models and time series forecasting models, such as ARIMA, as hybrid models (DT-ARIMA, RF-ARIMA) outperform the other models (Extra Tree Regressor, Random Forest, and Decision Tree). As compared to other models, Ridge Regression exhibits larger MSE, MAE, RMSE, and MAPE, as well as a negative R^2 value (–2.157), indicating poor generalization. This can be attributed to the model's linear assumption, which may not capture the complex, nonlinear relationships in the dataset, and potential multicollinearity among features. Inadequate regularization parameter tuning may also have contributed to the underperformance.

Table 8 Performance of TSRF for classification and regression combination

TSRF (Classification)		TSRF (Regression)	
Accuracy	0.9996	**MSE**	1.1780
Precision	0.9996	**MAE**	1.0710
Recall	0.9996	**MAPE**	1.2960
F1-Score	0.9997	**R2**	0.9680
Overall Accuracy Hybrid TSRF	0.9830		

The performance of the ANN is moderate, exhibiting neither exceptional nor poor results in its application. More data may be needed to capture the underlying patterns properly, or perhaps optimisation of the network architecture or hyperparameters is necessary. Particularly when it comes to MAPE, the time series data-specific ARIMA model performs better. It proves that ARIMA is successfully capturing the data's substantial temporal component. Hybrid models outperform ensemble approaches, such as Random Forest and Extra Tree Regressor, which nevertheless demonstrate high performance. It indicates that although ensemble approaches are effective on their own, they achieve even greater success when combined with time series models, as seen in RF-ARIMA.

Table 8 provides an overview of the performance metrics associated with the application of a TSRF approach in both classification and regression tasks. In the domain of classification, the TSRF model has exceptional performance across established measures, including accuracy, precision, recall, and F1-score, which are close to 0.9996. This suggests that the model is extremely dependable in accurately detecting and categorising the outcomes. The F1-score, with 0.9997, indicates a harmonious balance between precision and recall. This suggests that the model has been effectively calibrated to strike a balance between capturing all pertinent occurrences and ensuring a high degree of accuracy in its predictions. Nevertheless, the hybrid TSRF, which integrates classification and regression techniques, exhibits a marginal decrease in overall accuracy, reaching a value of 0.983. The slight decline, although insignificant, may indicate the intricacies and possible trade-offs associated with integrating classification and regression tasks.

With a MSE of 1.1780, the TSRF outperforms all models listed, including the DT-ARIMA (2.5260) and RF-ARIMA (2.9070), which were previously among the best performers. Similarly, the TSRF's mean absolute error (MAE) of 1.0710 is considerably lower than that of the next best models, such

as ARIMA (1.1390) and Decision Tree (1.4800). This indicates that the TSRF model provides more precise predictions compared to these individual models. The MAPE of TSRF is 1.2960, again outperforming other models where the lowest MAPE in Table 6 is 1.4330 for ARIMA. This highlights the TSRF's strength in minimizing the percentage error, making it a more reliable model for real-world applications where the accuracy of predictions relative to the actual values is crucial. Most importantly, the R^2 value of 0.9680 suggests that TSRF explains more variance in the data compared to other models. For instance, ARIMA has an R^2 of 0.8710, while even the hybrid models like DT-ARIMA and RF-ARIMA have R^2 values of 0.9320 and 0.9220, respectively. This shows that TSRF not only offers better overall accuracy but also captures the underlying patterns in the data more effectively. The k-fold cross-validation results for TSRF yield an average MSE of 1.1900 and R^2 of 0.9670, which are consistent with the results from the single 70/30 split (MSE = 1.1780, R^2 = 0.9680). The similarity in performance across folds confirms the model's robustness and generalization capability on unseen data, even without the use of external datasets.

In the context of WQI forecasting, error thresholds from previous studies (Lokman et al. 2025b) and operational standards from Malaysian DOE suggest that MSE values below 3.0 and MAE values below 1.5 are considered acceptable for decision-making purposes. These thresholds correspond to less than ± 5% deviation from laboratory-measured WQI values, ensuring predictions remain reliable for environmental management. Based on these benchmarks, TSRF's MSE of 1.1780 and MAE of 1.0710 fall well within the acceptable range, indicating high predictive precision.

To quantify the improvement of the proposed TSRF model over traditional approaches, its performance was compared with ARIMA, which was the best-performing conventional model. The TSRF model achieves a 75.6% reduction in MSE (from 4.8300 to 1.1780) and a 9.7% absolute increase in R^2 (from 0.8710 to 0.9680). These improvements indicate that TSRF can capture both temporal dependencies and complex non-linear relationships more effectively than ARIMA. Traditional models such as ARIMA and Ridge Regression are limited by their linear assumptions and their inability to fully capture multi-parameter interactions. In contrast, TSRF integrates decision tree regression for non-linear pattern learning, ARIMA for temporal trend modelling, and random forest classification for WQI category prediction. This hybrid approach enables TSRF to achieve higher predictive accuracy while maintaining interpretability for environmental water quality forecasting.

In addition to performance metrics, a residual versus fitted values plot was used to assess the reliability and potential overfitting of the TSRF model. The diagnostic tool helps

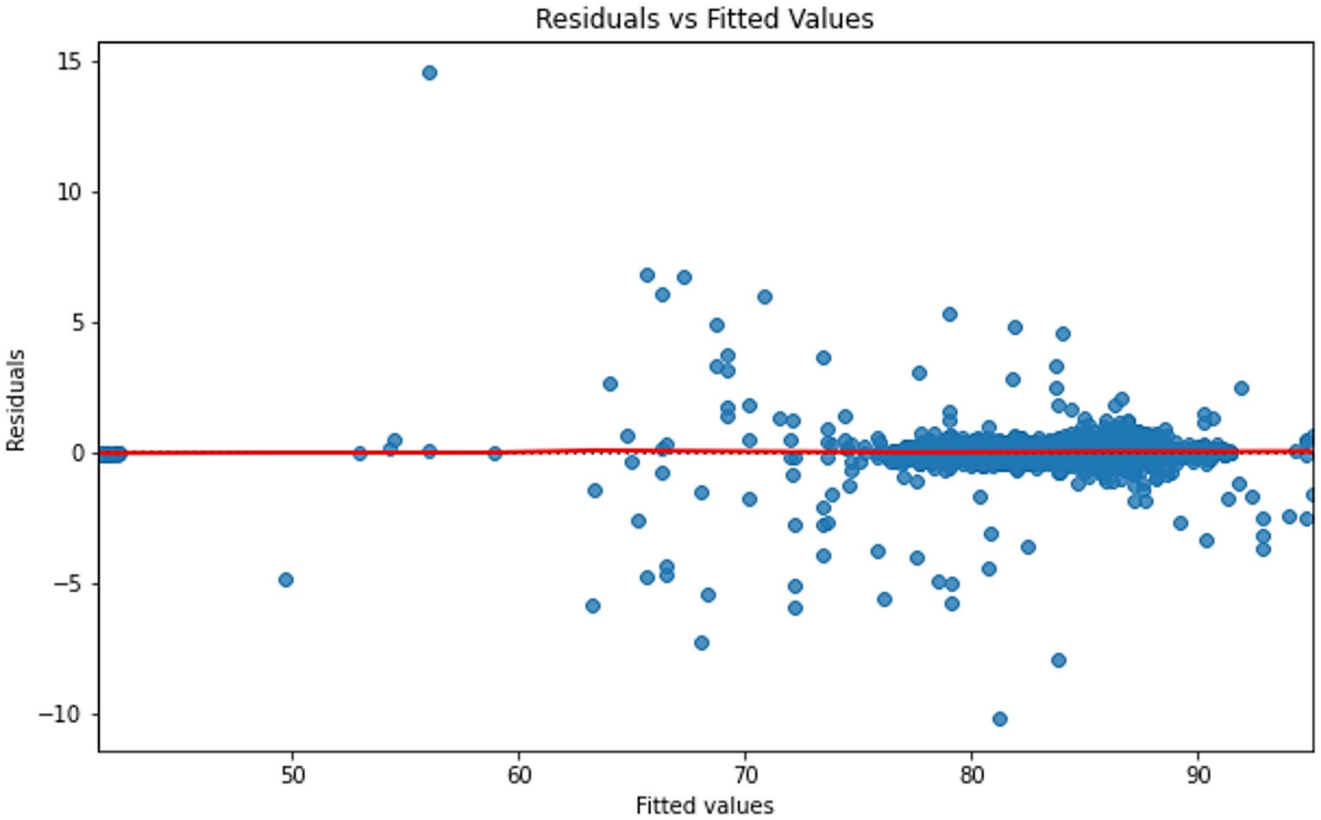


Fig. 8 Residuals vs. fitted values for the TSRF model

Table 9 Performance comparison of other predictive models for water quality forecasting based on data in Malaysia

Model	MSE $(mg/L)^2$	RMSE (mg/L)	MAE (mg/L)	MAPE (%)	R^2
LSTM	9.649	3.106	2.023	2.59%	0.879
GRU	24.873	4.987	2.175	4.02%	0.781
CNN	16.239	4.029	3.420	4.23%	0.796
CNN-LSTM	13.224	3.636	2.135	3.11%	0.834
Transformer	24.956	4.995	2.574	4.36%	0.781
Wavelet-ARIMA	36.874	6.072	2.512	4.76%	0.676
TSRF	1.178	1.085	1.070	1.29%	0.983

determine whether the residuals exhibit random behaviour or systematic patterns that might indicate model bias. As shown in Fig. 8, the residuals are scattered evenly around the zero line, without any clear trend or funnel-shaped pattern. This suggests that the variance of residuals remains relatively constant across predicted values, confirming the homoscedasticity assumption. Furthermore, the absence of structure or curvature in the residuals indicates that the model captures the underlying data behaviour well and does not overfit.

The comparative analysis in Table 9 shows that LSTM achieves MSE = 9.649, RMSE = 3.106, MAE = 2.023, MAPE = 2.59% and $R^2 = 0.879$, indicating strong predictive accuracy and reliability. CNN-LSTM also performs well, balancing low error metrics and good R^2 (0.834), while CNN achieves moderate accuracy but higher error rates compared to other models. In contrast, GRU and Transformer demonstrate relatively weaker performance, both requiring higher computational costs with lower accuracy ($R^2 = 0.781$). The Wavelet-ARIMA model shows the least effective prediction capability, reflected in the highest error metrics and lowest R^2 (0.676). Overall, the results suggest that deep learning models, particularly LSTM and CNN-LSTM, are more suitable for dynamic water quality forecasting compared to traditional hybrid approaches.

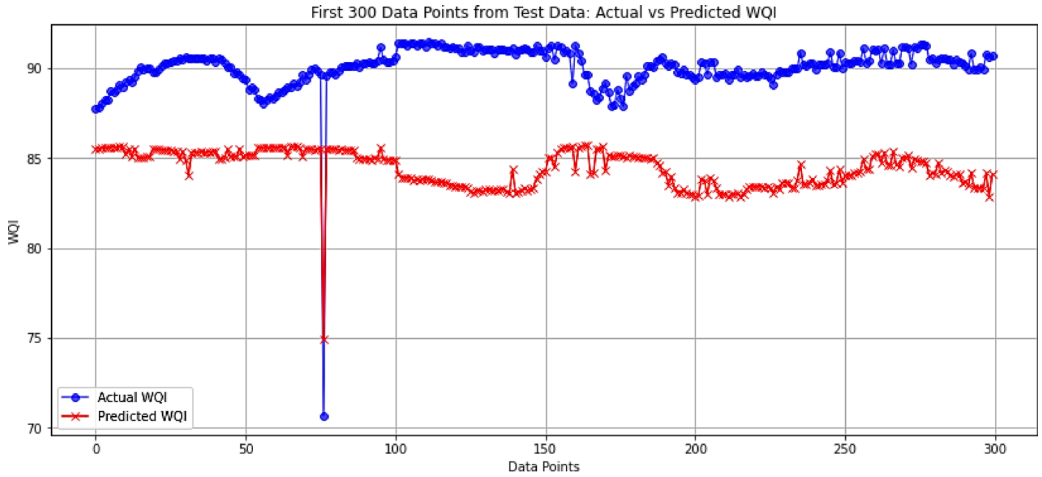
Beyond the models evaluated in this study, recent works have reported the use of advanced deep learning architectures such as CNN-BiLSTM (Zhang et al. 2022a, b), Wavelet-CNN (Wu and Wang 2022), and attention-based LSTM frameworks (Pang et al. 2024) for water quality forecasting. While these models have achieved notable accuracy improvements over conventional methods, their reported R^2 values generally range between 0.85 and 0.95, with error metrics remaining higher than those achieved by TSRF. In contrast, TSRF attains an R^2 of 0.983 and substantially lower MSE, RMSE, and MAPE values, highlighting its superior predictive performance. This suggests that TSRF not only outperforms the deep learning models tested in this study but also demonstrates competitive or even superior accuracy compared to other state-of-the-art deep learning approaches documented in the literature.

Figure 9 shows the actual and predicted values for each model. The graph presents a comparison between the actual and predicted WQI across each model. For clarity and detailed pattern observation, two graphs are depicted per model. The first graph captures the initial 300 test data points, offering a zoomed-in view to discern the trends more clearly, while the second graph encompasses the entire test dataset, constituting 30% of the total data, providing a comprehensive overview of the test data's behaviour

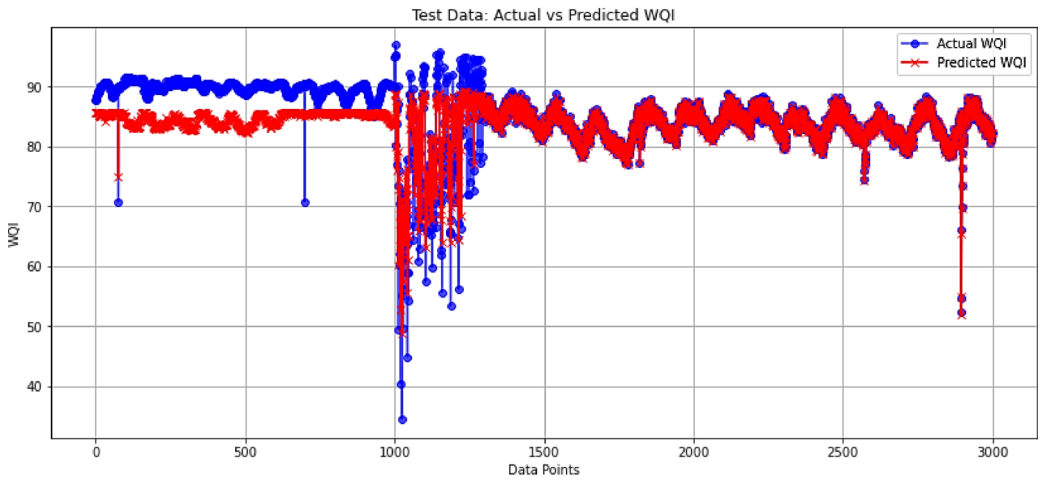
The ETR shows that the predicted values are tightly clustered around the actual data points. It shows that the model is strong and can understand complicated patterns in the dataset. Since its thorough approach to feature splits, this model usually performs well with different datasets. It is possible to see a moderate spread of points in the Ridge Regression plot. In cases when there is multicollinearity across features, this model performs exceptionally well because it provides a penalty that simplifies the model, which in turn helps to avoid overfitting and achieve a balanced pattern of predictions. Assuming it maintains a reasonable level of variance and bias, the RF should provide predictions that are quite near to the true values.

This research suggests that if the ANN prediction is very scattered, it may indicate the need for adjusting the model's settings or providing it with more learning opportunities to better learn from the data without memorizing it. Because of its proficiency in time-series analysis, the ARIMA model should show good agreement with observed values. To fully utilise time-based dependencies, this model works best when there are obvious trends or seasonal patterns in the data. A more obvious trend in forecasts might be shown using the Decision Tree graph. A wider distribution of points may indicate that this model has overfitted and failed to adequately represent the subtleties of the training data. Lastly, TSRF, DT-ARIMA and RF-ARIMA are expected to demonstrate a remarkable combination of predictions. For datasets that heavily rely on temporal patterns, DT-ARIMA and RF-ARIMA bring together the interpretability of Decision Trees or Random Forests with the time-series expertise of ARIMA, resulting in a dense clustering of predictions.

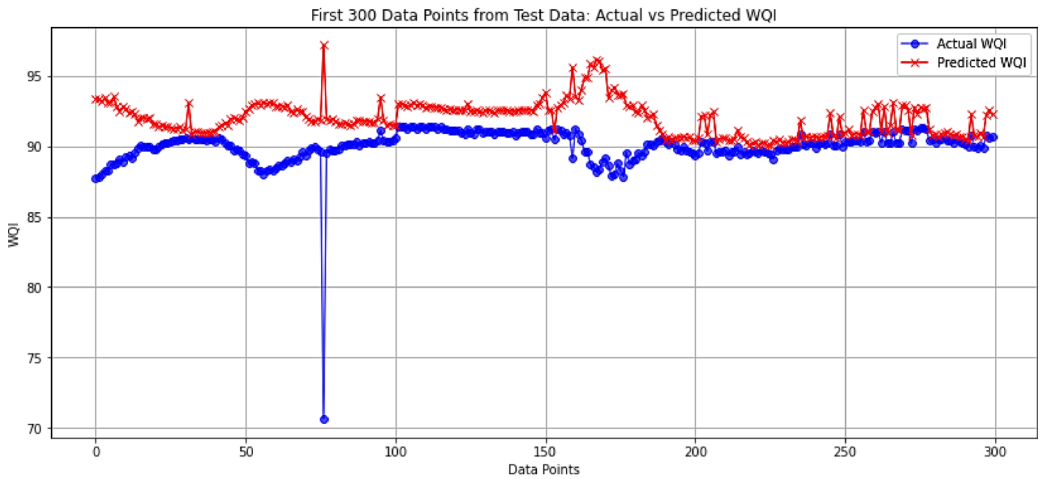
The proposed model, TSRF shows a better output than the other models. In comparison to other models, the TSRF model stands out for its remarkable capacity to replicate real WQI values, a sign of its resilience and output precision. This success is because TSRF is very good at dealing with time-dependent data, which is frequently marked by



(A) Extra Tree Regressor for first 300 test data points.

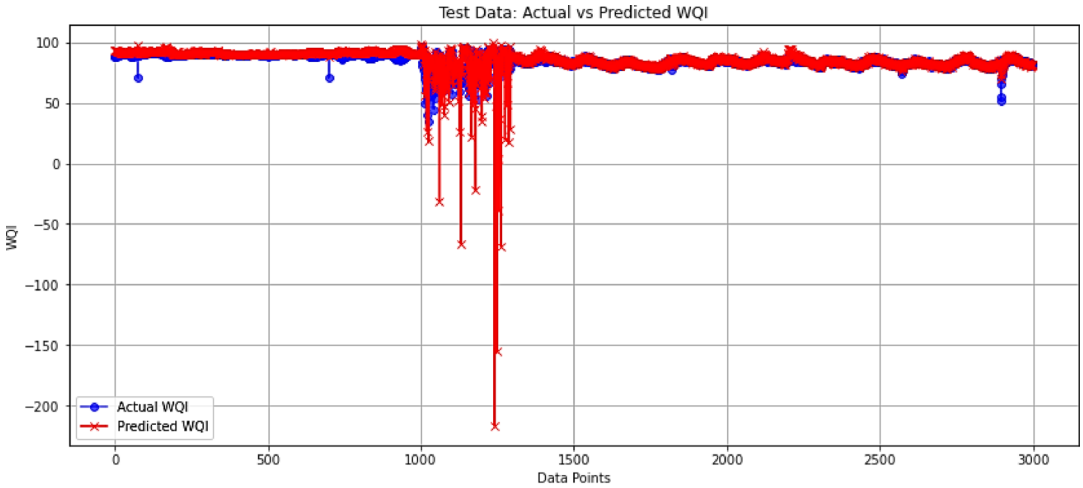


(B) Extra Tree Regressor for all test data points.

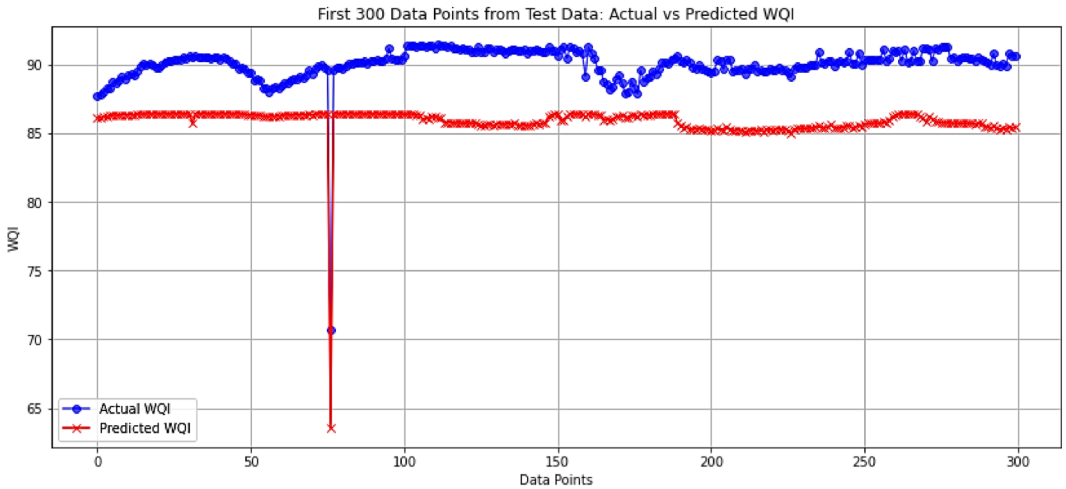


(C) Ridge Regression for first 300 test data points.

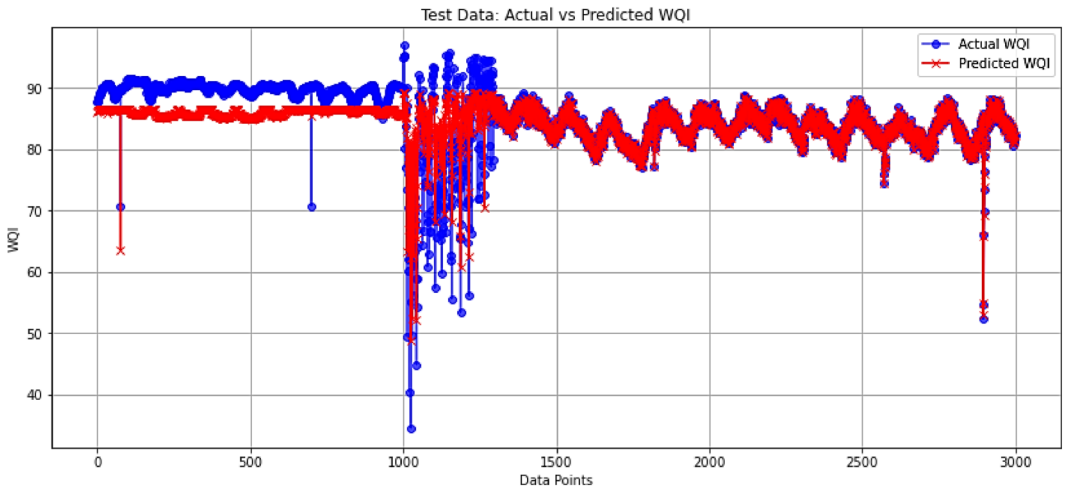
Fig. 9 **A**–**R** shows the actual and predicted values for various machine learning models



(D) Ridge Regression for all test data points.

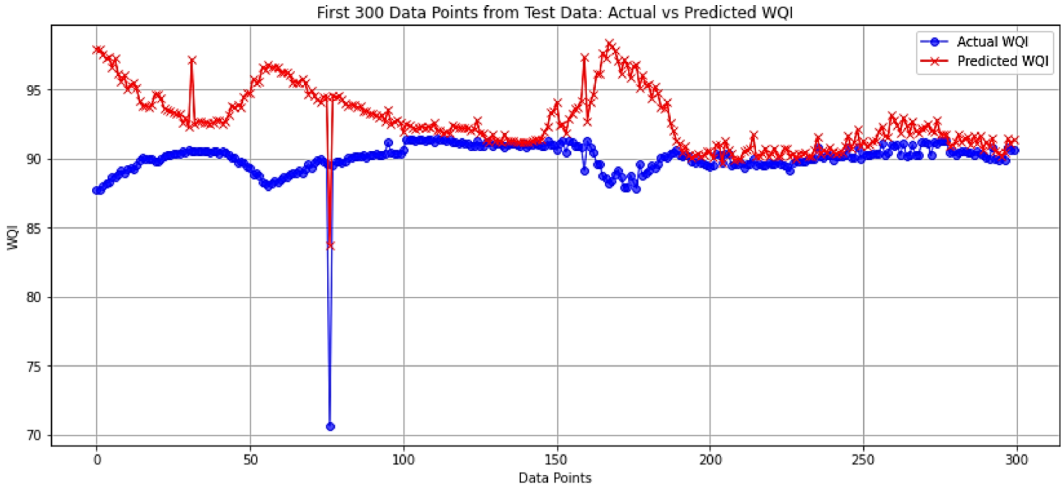


(E) Random Forest for first 300 test data points.

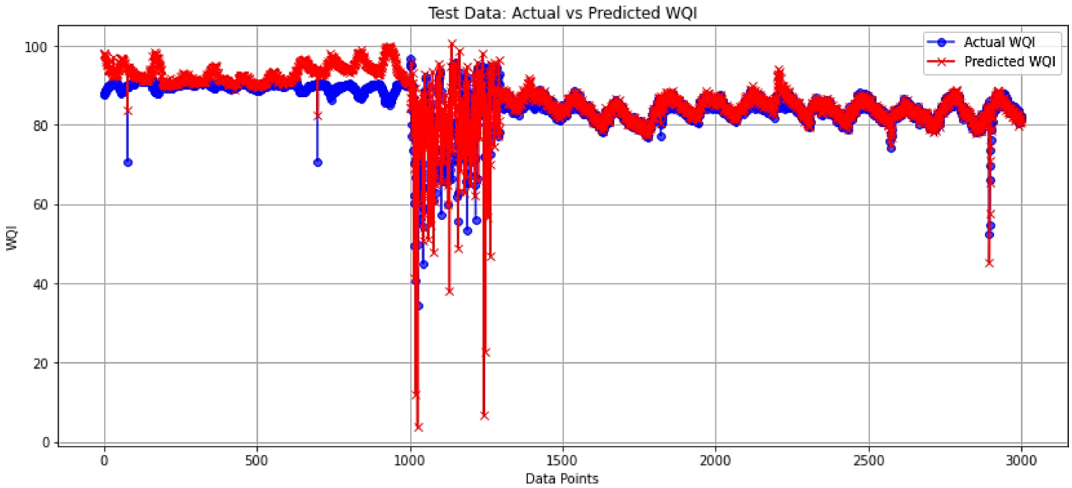


(F) Random Forest for all test data points.

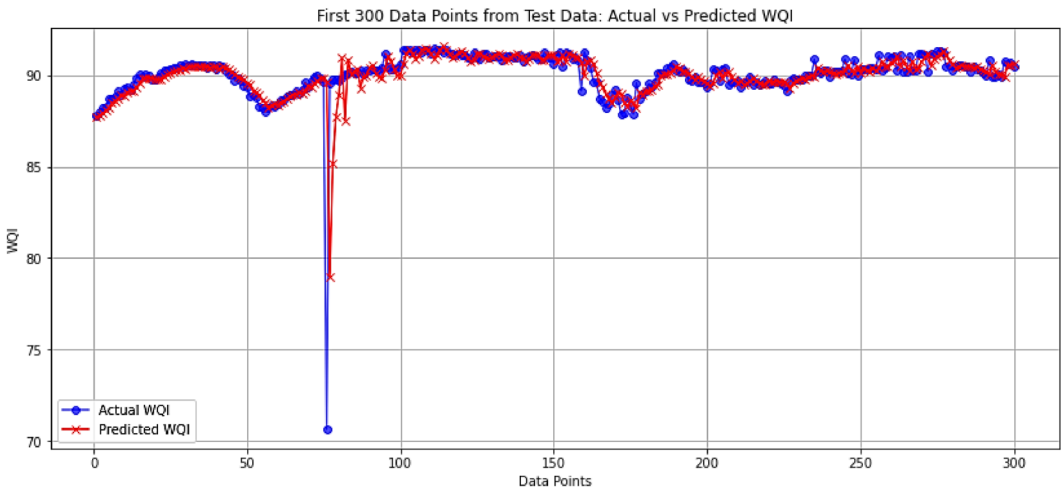
Fig. 9 (continued)



(G) ANN for first 300 test data points.

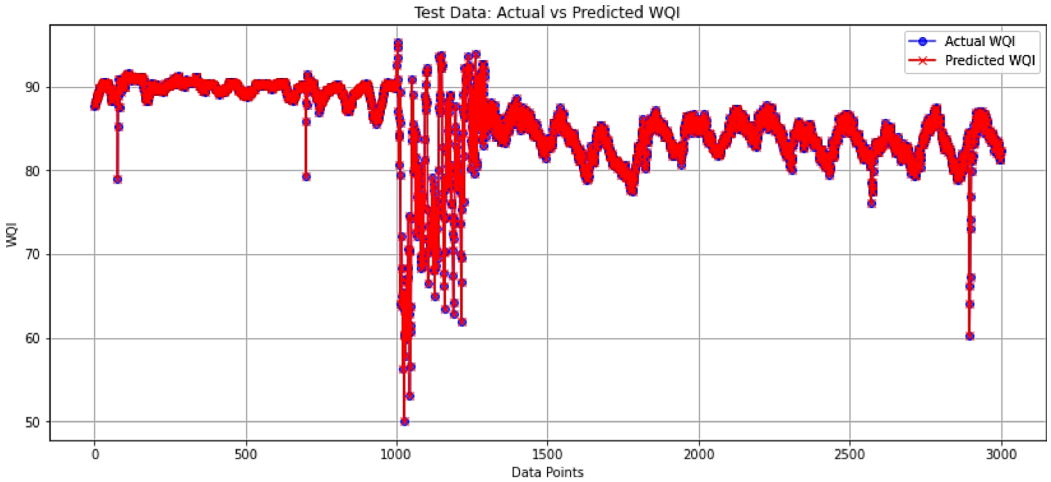


(H) ANN for all test data points.

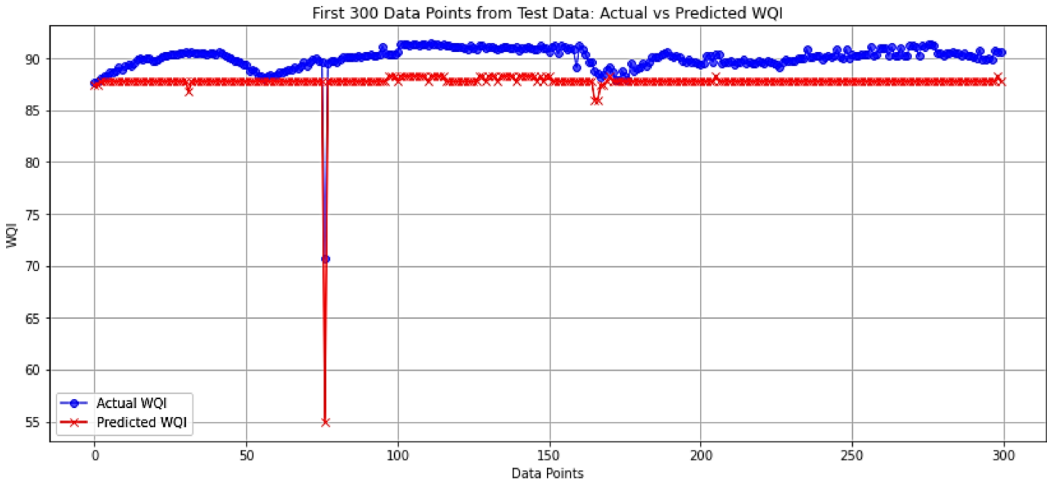


(I) ARIMA for first 300 test data points.

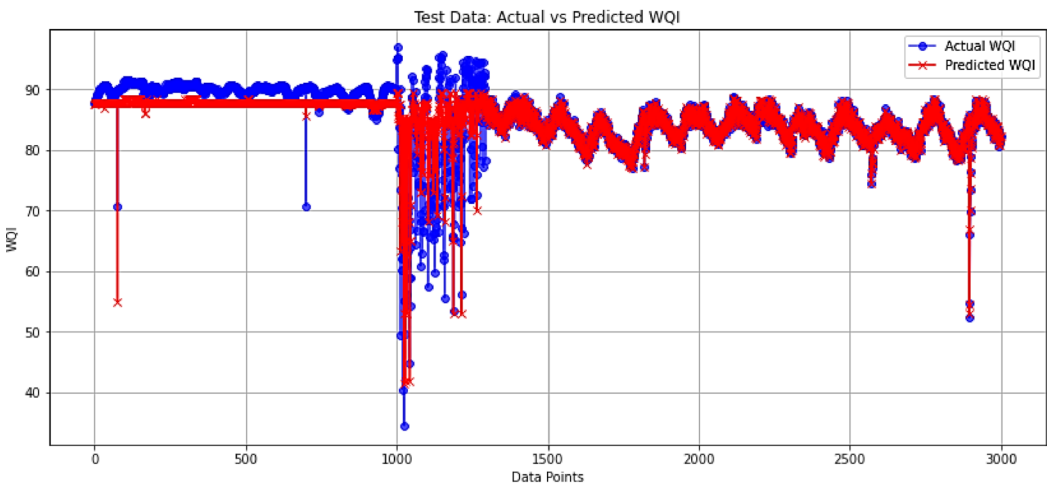
Fig. 9 (continued)



(J) ARIMA for all test data points.

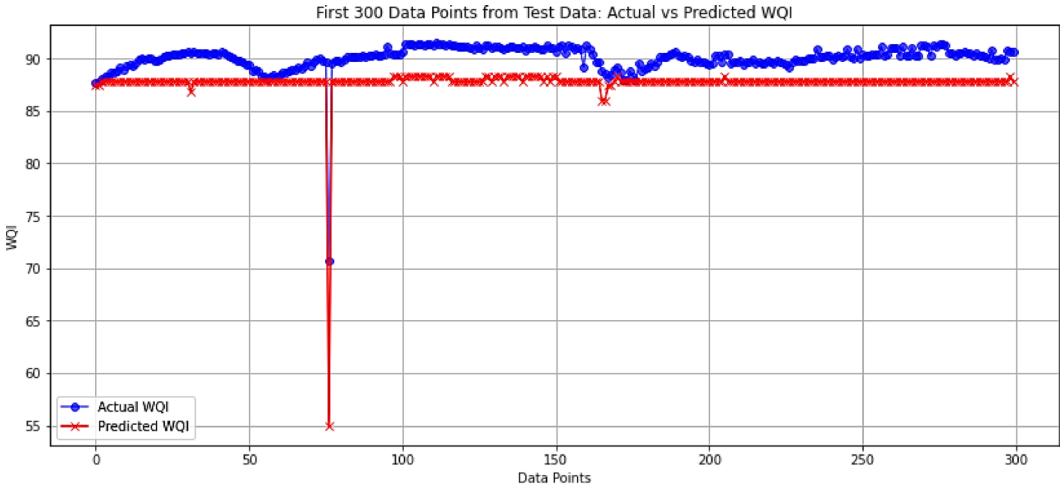


(K) Decision Tree for first 300 test data points.

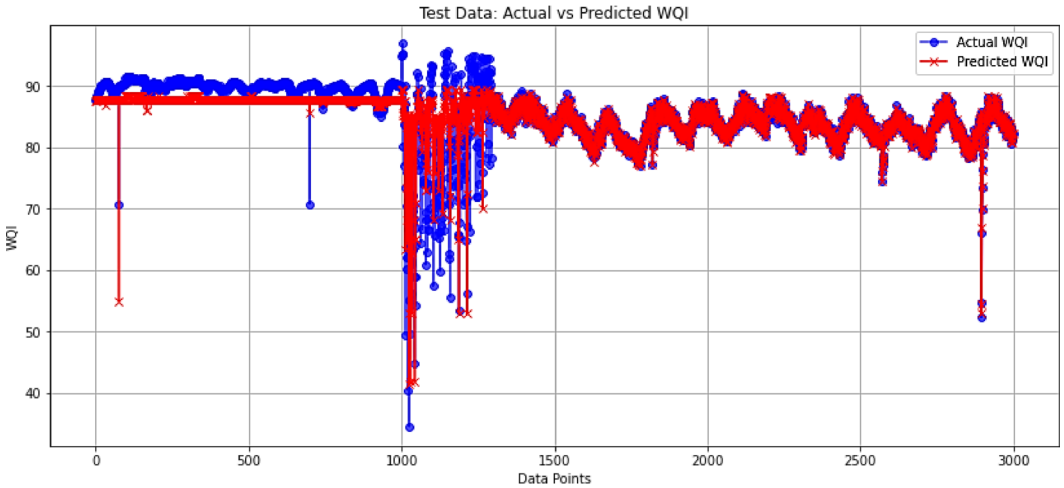


(L) Decision Tree for all test data points.

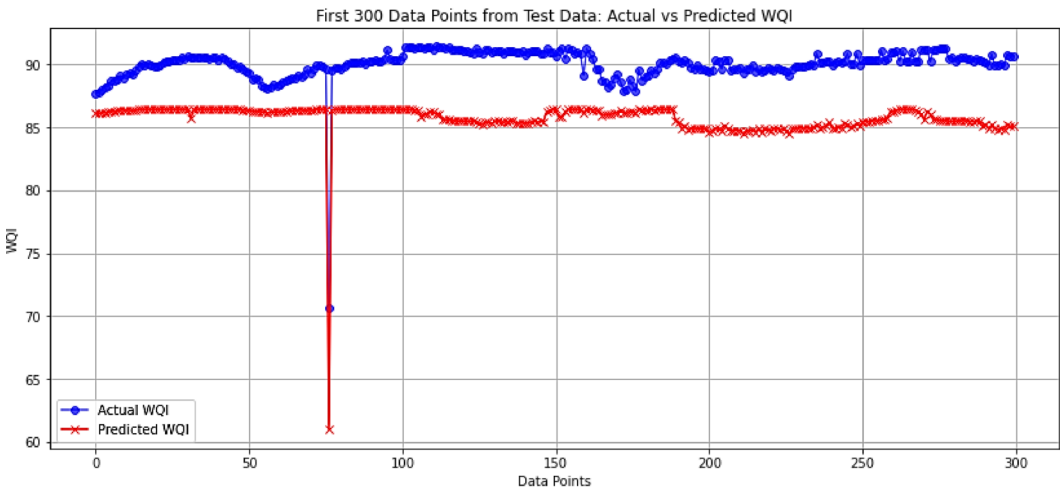
Fig. 9 (continued)



(M) DT-ARIMA for first 300 test data points.

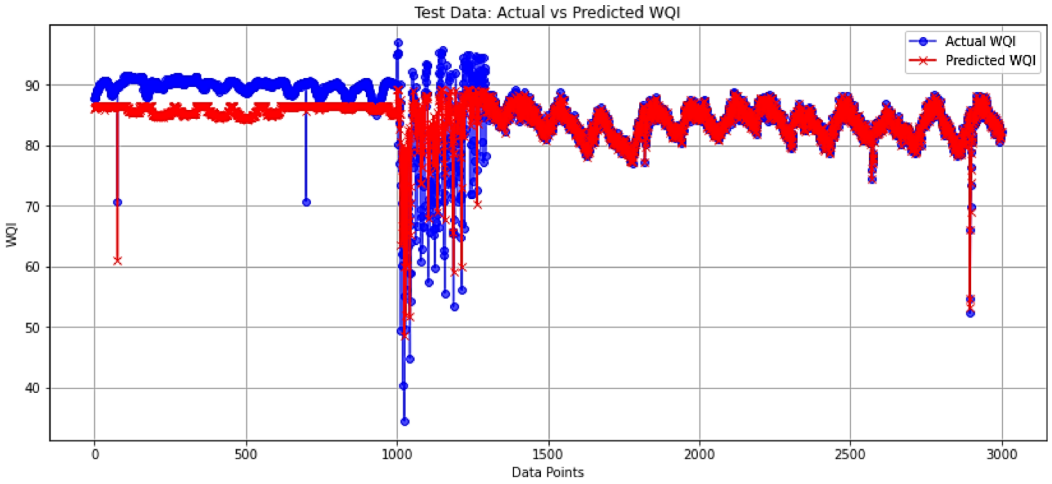


(N) DT-ARIMA for all test data points.

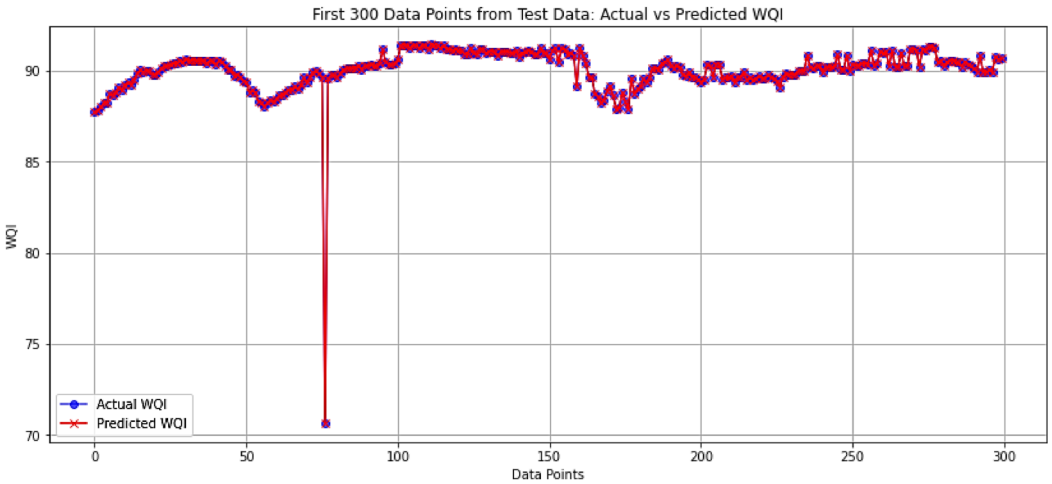


(O) DT-RF for first 300 test data points.

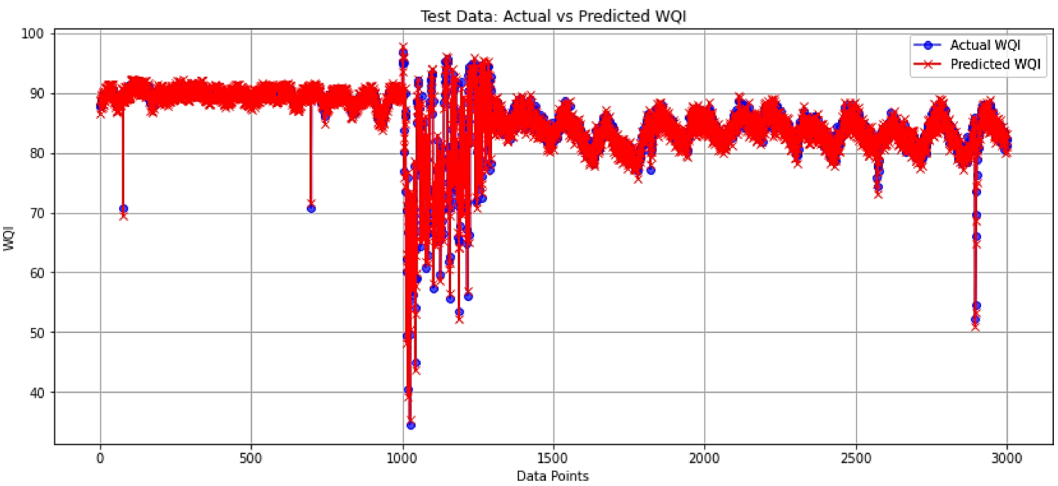
Fig. 9 (continued)



(P) DT-RF for all test data points.



(Q) TSRF for first 300 test data points.



(R) TSRF for all test data points.

Fig. 9 (continued)

intricate seasonal trends and patterns that more simplistic models may miss. The temporal dynamics are better captured by TSRF because it uses an ensemble of regression trees, each trained on separate parts of the time series data. Overfitting is a typical problem in time series forecasting; however, this ensemble method improves predicted accuracy by lowering variance. If the anticipated values fall within specific ranges or thresholds that define classes of water quality, TSRF can categorise data points into discrete categories as well as predict continuous output values like the WQI.

Conclusion

In conclusion, the TSRF model is an innovative method in predictive modelling, effectively combining the predictive capabilities of machine learning with the temporal accuracy of time series forecasting. This novel model integrates the precise data segmentation provided by decision trees with the sophisticated trend analysis capabilities of ARIMA models, resulting in an integrated solution that outperforms conventional models in terms of both breadth and precision. The TSRF prominently includes a Random Forest model, which serves as the foundation of the WQI class system. This feature leverages the ensemble learning technique of Random Forest to provide extremely accurate predictions of water quality classes, making it an essential element in the TSRF design. The Random Forest model is highly versatile and can handle complex and diverse data well, ensuring that TSRF not only delivers accurate continuous value forecasts but also offers precise insights into discrete classes. As we consider the future, refining the TSRF model will require a thorough examination of prediction inaccuracies and a focused effort to understand the importance and interaction of different characteristics. These efforts will lead to improved model performance while maintaining interpretability, especially in contexts that require clarity in algorithmic decision-making. It is essential to validate the robustness of TSRF using new data to ensure its reliability and durability in the face of changing environmental conditions, and to adopt a proactive strategy for updating models in line with emerging data patterns.

From an application standpoint, TSRF can be deployed for real-time water quality monitoring in Malaysian river systems, such as the Klang River, to provide continuous WQI forecasts for pollution control and public safety alerts. Its predictive capabilities can also be integrated into the Malaysia Water Sector Transformation 2040 framework, supporting proactive resource management, early warning systems, and data-driven policymaking. These integrations will enable government agencies, water authorities, and environmental stakeholders to act swiftly on predictive insights, translating technical performance into tangible environmental and public health benefits. TSRF demonstrates the advanced complexity and promise of hybrid modelling techniques in analysing environmental data due to its integrated structure and classification capabilities.

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Data availability All data can be requested from DOE, Malaysia.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval Not applicable.

Consent for publication All authors agreed to publish the manuscript.

Consent to participate Not applicable.

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